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Quality Assurance (IM353IA): Unit I

Department of Industrial Engineering and Management

Go, change the world

Unit I

Introduction to Quality | Statistical Process Control

Introduction to Statistical Quality Control: Quality Improvement in the Modern Business Environment

Introduction to Statistical Quality Control, 5th edition

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1 Quality Improvement in the Modern Business Environment

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CHAPTER OUTLINE

1-1 THE MEANING OF QUALITY AND QUALITY IMPROVEMENT

- 1-1.1 Dimensions of Quality
- 1-1.2 Quality Engineering Terminology

1-2 A BRIEF HISTORY OF QUALITY CONTROL AND IMPROVEMENT

1-3 STATISTICAL METHODS FOR QUALITY CONTROL AND IMPROVEMENT

1-4 MANAGEMENT ASPECTS OF QUALITY IMPROVEMENT

- 1-4.1 Quality Philosophy and Management Strategies
- 1-4.2 The Link Between Quality and Productivity
- 1-4.3 Quality Costs
- 1-4.4 Legal Aspects of Quality
- 1-4.5 Implementing Quality Improvement

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Learning Objectives

1. Define and discuss quality and quality improvement
2. Discuss the different dimensions of quality
3. Discuss the evolution of modern quality improvement methods
4. Discuss the role that variability and statistical methods play in controlling and improving quality
5. Describe the quality management philosophies of W. Edwards Deming, Joseph M. Juran, and Armand V. Feigenbaum
6. Discuss total quality management, the Malcolm Baldrige National Quality Award, Six-Sigma, and quality systems and standards
7. Explain the links between quality and productivity and between quality and cost
8. Discuss product liability
9. Discuss the three functions: quality planning, quality assurance, and quality control and improvement

1-1 Definitions and Meaning of Quality

1-1.1 The Eight Dimensions of Quality

1. Performance
2. Reliability
3. Durability
4. Serviceability
5. Aesthetics
6. Features
7. Perceived Quality
8. Conformance to Standards

Definition

Quality means fitness for use.

- This is a traditional definition
- Quality of design
- Quality of conformance

Definition

Quality is inversely proportional to variability.

This is a modern definition of quality

The Transmission Example

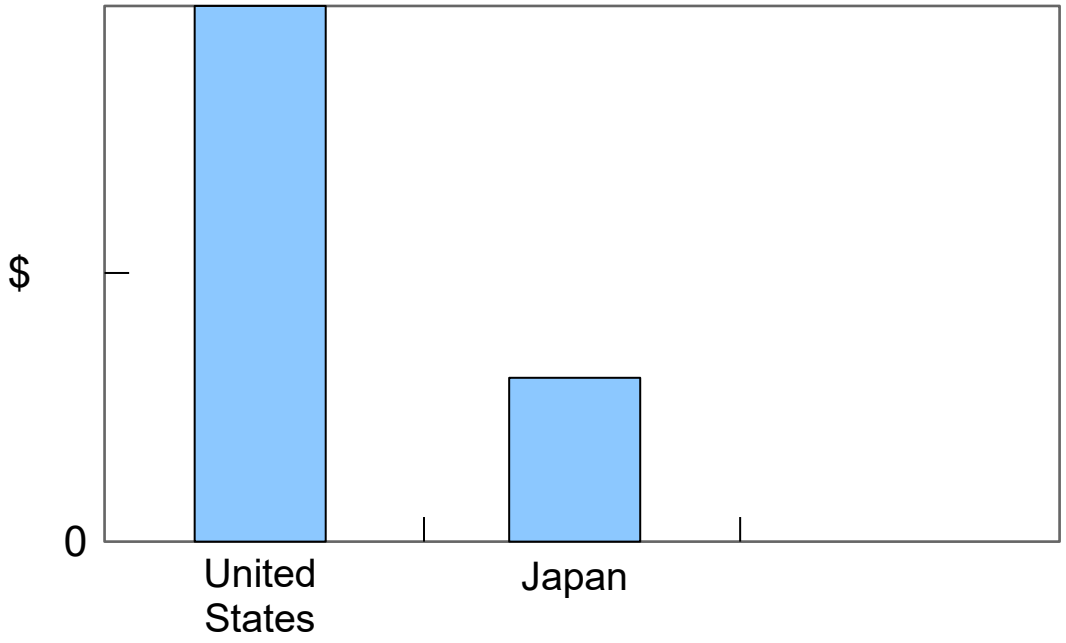


Figure 1-1 Warranty costs for transmissions.

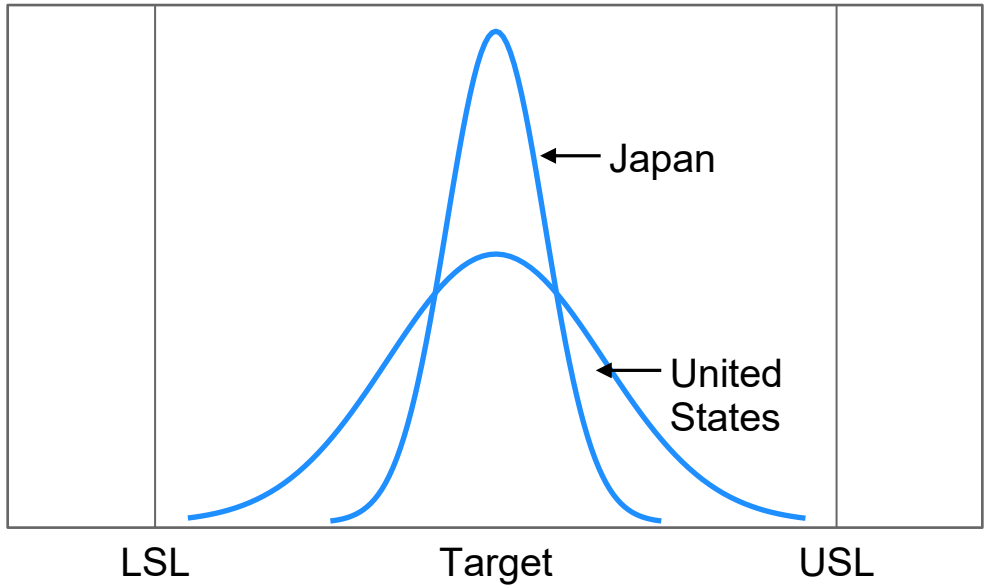


Figure 1-2 Distributions of critical dimensions for transmissions.

Definition

Quality improvement is the reduction of variability in processes and products.

- The transmission example illustrates the utility of this definition
- An equivalent definition is that quality improvement is the **elimination of waste**. This is useful in service or transactional businesses.

1-1.2 Terminology

Every product possesses a number of elements that jointly describe what the user or consumer thinks of as quality. These parameters are often called **quality characteristics**. Sometimes these are called **critical-to-quality (CTQ)** characteristics. Quality characteristics may be of several types:

1. **Physical:** length, weight, voltage, viscosity
2. **Sensory:** taste, appearance, color
3. **Time Orientation:** reliability, durability, serviceability

Since variability can only be described in statistical terms, **statistical methods** play a central role in quality improvement efforts. In the application of statistical methods to quality engineering, it is fairly typical to classify data on quality characteristics as either **attributes** or **variables** data. Variables data are usually continuous measurements, such as length, voltage, or viscosity. Attributes data, on the other hand, are usually discrete data, often taking the form of counts. We will describe statistical-based quality engineering tools for dealing with both types of data.

Terminology cont'd

- Specifications
  - Lower specification limit
  - Upper specification limit
  - Target or nominal values
- Defective or nonconforming product
- Defect or nonconformity
- Not all products containing a defect are necessarily defective

1-2. History of Quality Improvement

Table 1-1 A Timeline of Quality Methods

1700–1900	Quality is largely determined by the efforts of an individual craftsman. Eli Whitney introduces standardized, interchangeable parts to simplify assembly.
1875	Frederick W. Taylor introduces "Scientific Management" principles to divide work into smaller, more easily accomplished units—the first approach to dealing with more complex products and processes. The focus was on productivity. Later contributors were Henry Gilbreth and Frank Gantt.
1900–1930	Henry Ford—the assembly line—further refinement of work methods to improve productivity and quality; Ford developed mistake-proof assembly concepts, self-checking, and inprocess inspection.
1901	First standards laboratories established in Great Britain.
1907–1908	AT&T begins systematic inspection and testing of products and materials.
1908	W.S. Gosset (writing as "Student") introduces the t-distribution—results from his work on quality control at Guinness Brewery.
1915–1919	WWI—British government begins a supplier certification program.
1919	Technical Inspection Association is formed in England; this later becomes the Institute of Quality Assurance.

1-2. History of Quality Improvement (cont'd)

Table 1-1 A Timeline of Quality Methods

1920s	AT&T Bell Laboratories forms a quality department—emphasizing quality, inspection and test, and product reliability. B. P. Dudding at General Electric in England uses statistical methods to control the quality of electric lamps.
1922–1923	R.A. Fisher publishes series of fundamental papers on designed experiments and their application to the agricultural sciences.
1924	W.A. Shewhart introduces the control chart concept in a Bell Laboratories technical memorandum.
1928	Acceptance sampling methodology is developed and refined by H. F. Dodge and H. G. Romig at Bell Labs.
1931	W.A. Shewhart publishes *Economic Control of Quality of Manufactured Product*—outlining statistical methods for use in production and control chart methods.
1932	W.A. Shewhart gives lectures on statistical methods in production and control charts at the University of London.
1932–1933	British textile and woolen industry and German chemical industry begin use of designed experiments for product/process development.
1933	The Royal Statistical Society forms the Industrial and Agricultural Research Section.

1-2. History of Quality Improvement

1938	W.E. Deming invites Shewhart to present seminars on control charts at the U.S. Department of Agriculture Graduate School.
1940	The U.S. War Department publishes a guide for using control charts to analyze process data.
1940–1943	Bell Labs develop the forerunners of the military standard sampling plans for the U.S. Army.
1942	In Great Britain, the Ministry of Supply Advising Service on Statistical Methods and Quality Control is formed.
1942–1946	Training courses on statistical quality control are given to industry; more than 15 quality societies are formed in North America.
1944	*Industrial Quality Control* begins publication.
1946	The American Society for Quality Control (ASQC) is formed as the merger of various quality societies. The International standards organization (ISO) is founded. Deming is invited to Japan by the Economic and Scientific Services Section of the U.S. War Department to help occupation forces in rebuilding Japanese industry. The Japanese Union of Scientists and Engineers (JUSE) is formed.
1946–1949	Deming is invited to give statistical quality control seminars to Japanese industry.
1948	G. Taguchi begins study and application of experimental design.
1950	Deming begins education of Japanese industrial managers; statistical quality control methods begin to be widely taught in Japan. K. Ishikawa introduces the cause-and-effect diagram.
1950s	Classic texts on statistical quality control by Eugene Grant and A. J. Duncan appear.

1-2. History of Quality Improvement

1951	A. V. Feigenbaum publishes the first edition of his book, *Total Quality Control*. JUSE establishes the "Deming Prize" for significant achievement in quality control and quality methodology.
1951+	G. E. P. Box and K. B. Wilson publish fundamental work on using designed experiments and response surface methodology for process optimization; focus is on chemical industry. Applications of designed experiments in the chemical industry grow steadily after this.
1954	Joseph M. Juran is invited by the Japanese to lecture on quality management and improvement. British statistician E. S. Page introduces the cumulative sum (CUSUM) control chart.
1957	J. M. Juran and F. M. Gryna's *Quality Control Handbook* is first published.
1959	*Technometrics* (a journal of statistics for the physical, chemical, and engineering sciences) is established; J. Stuart Hunter is the founding editor. S. Roberts introduces the exponentially weighted moving average (EWMA) control chart. The U.S. manned spaceflight program makes industry aware of the need for reliable products; the field of reliability engineering grows from this starting point.
1960	G. E. P. Box and J. S. Hunter write fundamental papers on 2^{k-p} factorial designs. The quality control circle concept is introduced in Japan by K. Ishikawa.

1-2. History of Quality Improvement (cont'd)

1961	National Council for Quality and Productivity is formed in Great Britain as part of the British Productivity Council.
1960s	Courses in statistical quality control become widespread in Industrial Engineering academic programs. Zero defects (ZD) programs are introduced in certain U.S. industries.
1969	*Industrial Quality Control* ceases publication, replaced by *Quality Progress* and the *Journal of Quality Technology* (Lloyd S. Nelson is the founding editor of *JQT*).
1970s	In Great Britain the NCQP and the Institute of Quality Assurance merge to form the British Quality Association.
1975–1978	Books on designed experiments oriented toward engineers and scientists begin to appear. Interest in quality circles begins in North America—this grows into the total quality management (TQM) movement.

1-2. History of Quality Improvement

1980s	Experimental design methods are introduced to and adopted by a wider group of organizations, including electronics, aerospace, semiconductor, and the automotive industries. The works of Taguchi on designed experiments first appear in the United States.
1984	The American Statistical Association (ASA) establishes the Ad Hoc Committee on Quality and Productivity; this later becomes a full Section of the ASA. The journal *Quality and Reliability Engineering International* appears.
1986	Box and others visit Japan, noting the extensive use of designed experiments and other statistical methods.
1987	ISO publishes the first quality systems standard.
1988	The Malcolm Baldrige National Quality Award is established by the U.S. Congress. The European Foundation for Quality Management is founded; this organization administers the European Quality Award.
1989	The journal *Quality Engineering* appears. Motorola's six-sigma initiative begins.

1-2. History of Quality Improvement (cont'd)

1990s	ISO 9000 certification activities increase in U.S. industry; applicants for the Baldrige award grow steadily; many states sponsor quality awards based on the Baldrige criteria.
1995	Many undergraduate engineering programs require formal courses in statistical techniques, focusing on basic methods for process characterization and improvement.
1997	Motorola's six-sigma approach spreads to other industries.
1998	The American Society for Quality Control becomes the American Society for Quality (see www.asq.org), attempting to indicate the broader aspects of the quality improvement field.
2000s	ISO 9000:2000 standard is issued. Supply-chain management and supplier quality become even more critical factors in business success. Quality improvement activities expand beyond the traditional industrial setting into many other areas including financial services, health care, insurance, and utilities.

1-3 STATISTICAL METHODS FOR QUALITY CONTROL AND IMPROVEMENT

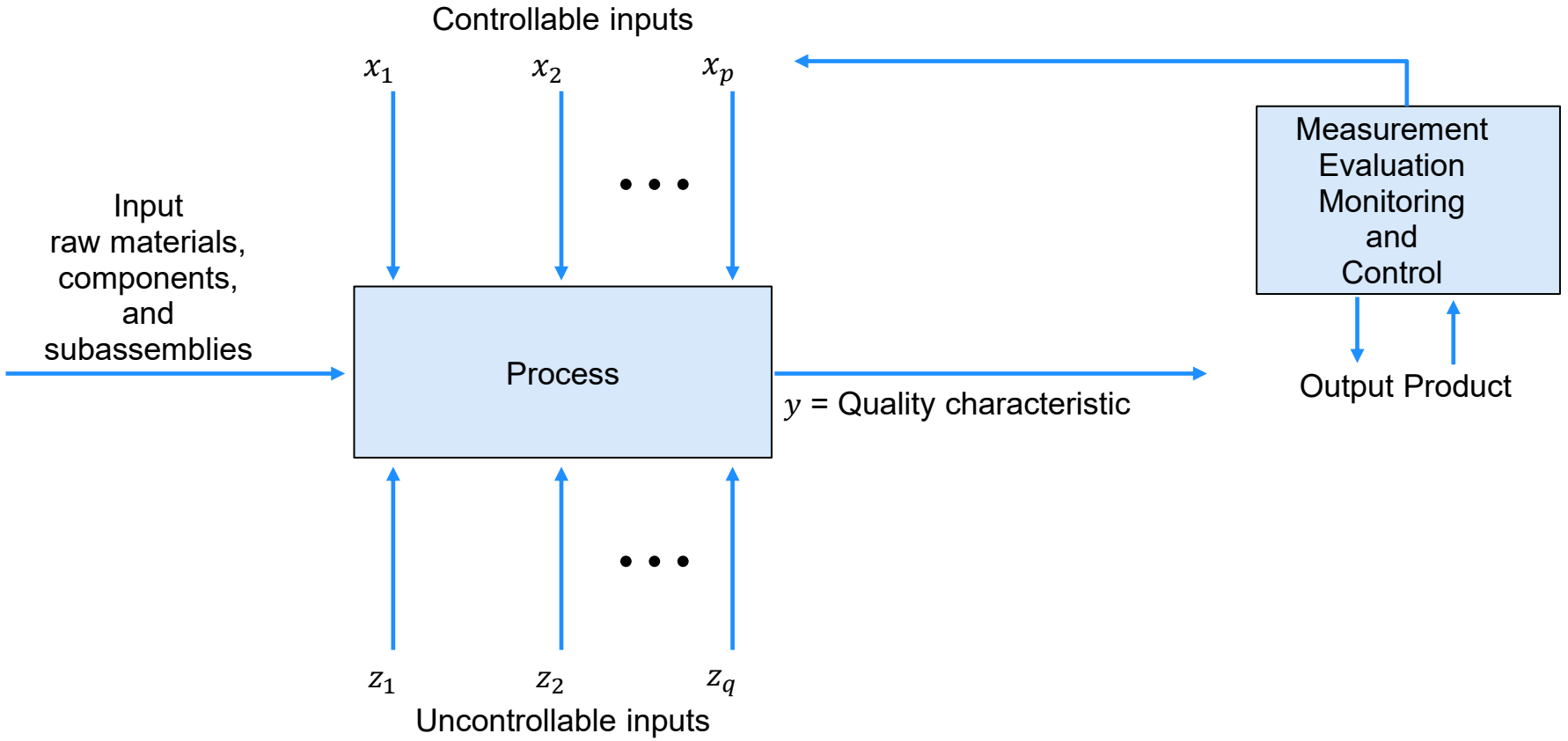
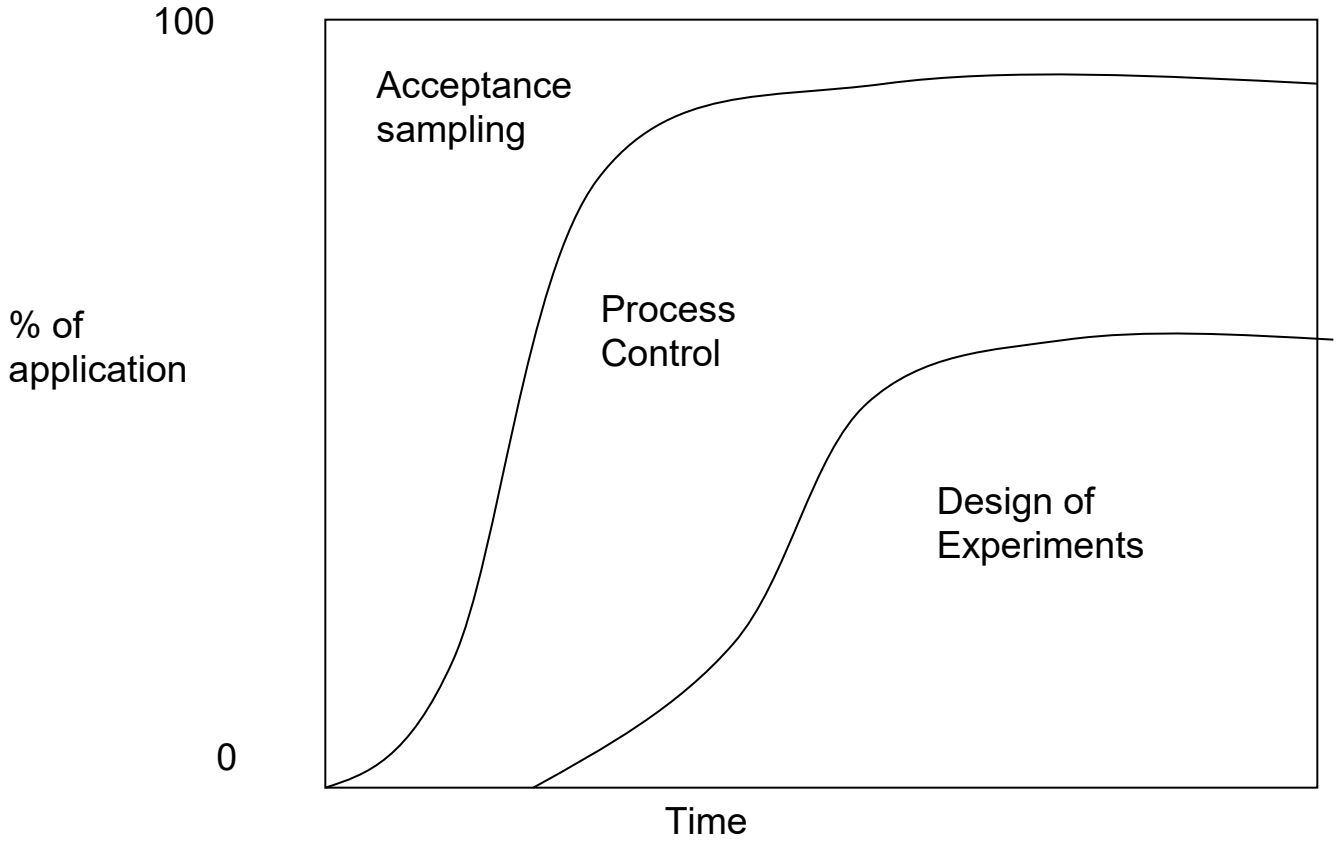


Figure 1-3 Production process inputs and outputs.

Statistical Methods

- Statistical process control (SPC)
  - Control charts, plus other problem-solving tools
  - Useful in monitoring processes, reducing variability through elimination of assignable causes
  - On-line technique
- Designed experiments (DOX)
  - Discovering the key factors that influence process performance
  - Process optimization
  - Off-line technique
- Acceptance Sampling

Phase diagram of the use of Quality Engineering methods.



Walter A. Shewart (1891-1967)

- Trained in engineering and physics
- Long career at Bell Labs
- Developed the first control chart about 1924

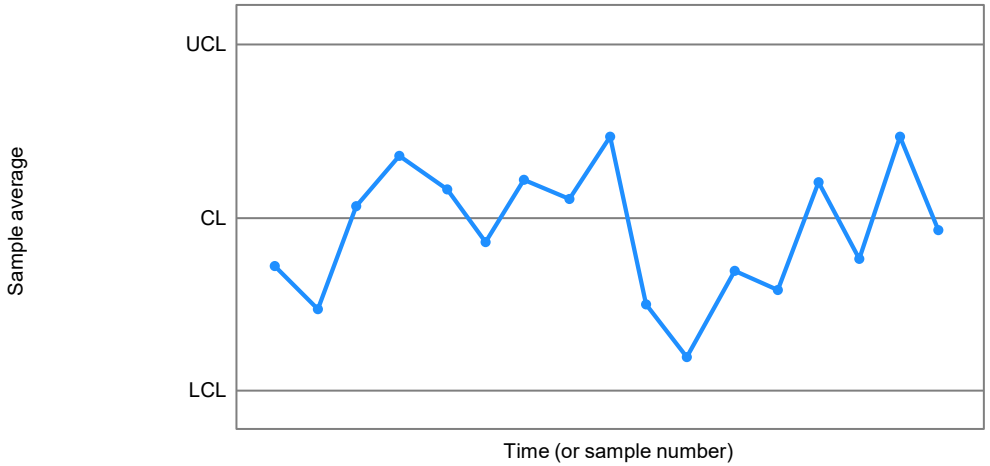
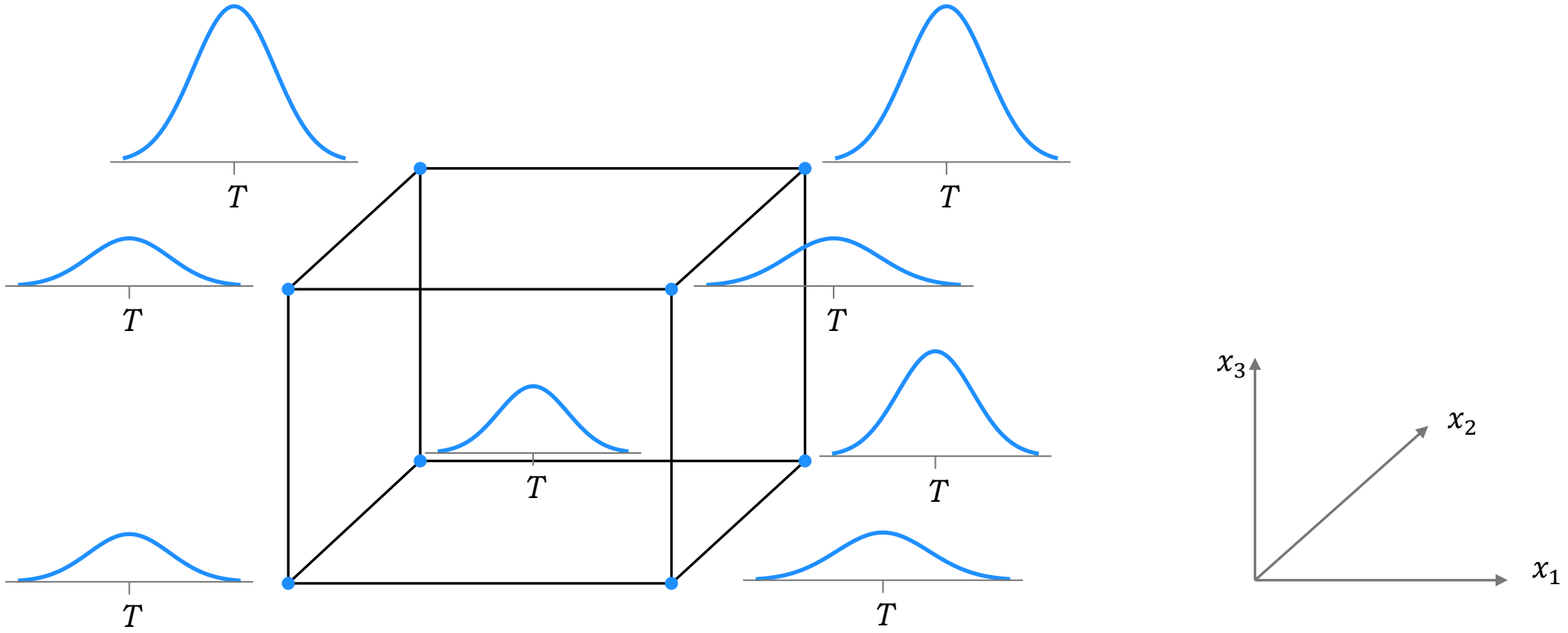


Figure 1-4 A typical control chart.





A factorial experiment with three factors

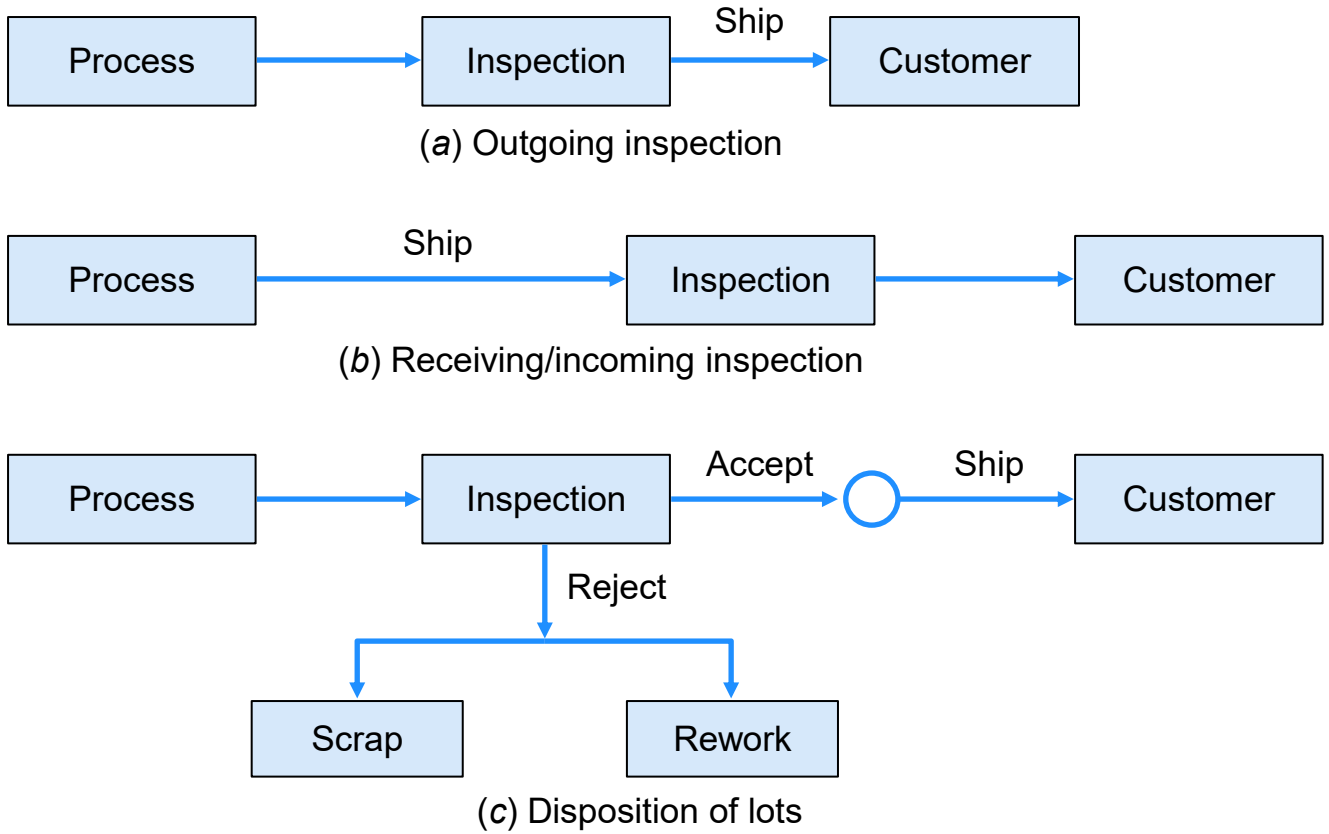


Figure 1-6 Variations of acceptance sampling.

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1-4 MANAGEMENT ASPECTS OF QUALITY IMPROVEMENT

Effective management of quality requires the execution of three activities:

1. Quality Planning
2. Quality Assurance
3. Quality Control and Improvement

Quality planning is a strategic activity, and it is just as vital to an organization's long-term business success as the product development plan, the financial plan, the marketing plan, and plans for the utilization of human resources. Without a strategic quality plan, an enormous amount of time, money, and effort will be wasted by the organization dealing with faulty designs, manufacturing defects, field failures, and customer complaints. Quality planning involves identifying customers, both external and those that operate internal to the business, and identifying their needs (this is sometimes called listening to the **voice of the customer**). Then products or services that meet or exceed customer expectations must be developed. The eight dimensions of quality discussed in Section 1-1.1 are an important part of this effort. The organization must then determine how these products and services will be realized. Planning for quality improvement on a specific, systematic basis is also a vital part of this process.

Quality assurance is the set of activities that ensures the quality levels of products and services are properly maintained and that supplier and customer quality issues are properly resolved. Documentation of the quality system is an important component. Quality system documentation involves four components: policy, procedures, work instructions and specifications, and records. Policy generally deals with what is to be done and why, while procedures focus on the methods and personnel that will implement policy. Work instructions and specifications are usually product-, department-, tool-, or machine-oriented. Records are a way of documenting the policies, procedures, and work instructions that have been followed. Records are also used to track specific units or batches of product, so that it can be determined exactly how they were produced. Records are often vital in providing data for dealing with customer complaints, corrective actions, and, if necessary, product recalls. Development, maintenance, and control of documentation are important quality assurance functions. One example of document control is ensuring that specifications and work instructions developed for operating personnel reflect the latest design and engineering changes.

Quality control and improvement involve the set of activities used to ensure that the products and services meet requirements and are improved on a continuous basis. Since variability is often a major source of poor quality, statistical techniques, including SPC and designed experiments, are the major tools of quality control and improvement. Quality improvement is often done on a project-by-project basis and involves teams led by personnel with specialized knowledge of statistical methods and experience in applying them. Projects should be selected so that they have significant business impact and are linked with the overall business goals for quality identified during the planning process. The techniques in this book are integral to successful quality control and improvement.

1-4.1 Quality Philosophies and Management Strategies

W. Edwards Deming

- Taught engineering, physics in the 1920s, finished PhD in 1928
- Met Walter Shewhart at Western Electric
- Long career in government statistics, USDA, Bureau of the Census
- During WWII, he worked with US defense contractors, deploying statistical methods
- Sent to Japan after WWII to work on the census



Deming

- Deming was asked by JUSE to lecture on statistical quality control to management
- Japanese adopted many aspects of Deming's management philosophy
- Deming stressed "continual never-ending improvement"
- Deming lectured widely in North America during the 1980s; he died 24 December 1993

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Deming's 14 Points

1. Create constancy of purpose toward improvement
2. Adopt a new philosophy, recognize that we are in a time of change, a new economic age
3. Cease reliance on mass inspection to improve quality
4. End the practice of awarding business on the basis of price alone
5. Improve constantly and forever the system of production and service
6. Institute training
7. Improve leadership, recognize that the aim of supervision is help people and equipment to do a better job
8. Drive out fear
9. Break down barriers between departments

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14 Points cont'd

10. Eliminate slogans and targets for the workforce such as zero defects
11. Eliminate work standards
12. Remove barriers that rob workers of the right to pride in the quality of their work
13. Institute a vigorous program of education and self-improvement
14. Put everyone to work to accomplish the transformation

Note that the 14 points are about change

Deming's Deadly Diseases

1. Lack of constancy of purpose
2. Emphasis on short-term profits
3. Performance evaluation, merit rating, annual reviews
4. Mobility of management
5. Running a company on visible figures alone
6. Excessive medical costs for employee health care
7. Excessive costs of warrantees

Joseph M. Juran

- Born in Romania (1904), immigrated to the US
- Worked at Western Electric, influenced by Walter Shewhart
- Emphasizes a more strategic and planning oriented approach to quality than does Deming
- Juran Institute is still an active organization promoting the Juran philosophy and quality improvement practices



The Juran Trilogy

1. Planning
2. Control
3. Improvement

- These three processes are interrelated
- Control versus breakthrough
- Project-by-project improvement

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Some of the Other "Gurus"

- Kaoru Ishikawa
  - Son of the founder of JUSE, promoted widespread use of basic tools
- Armand Feigenbaum
  - Author of Total Quality Control, promoted overall organizational involvement in quality,
  - Three-step approach emphasized quality leadership, quality technology, and organizational commitment
- Lesser gods, false prophets

Total Quality Management (TQM)

- Started in the early 1980s, Deming/Juran philosophy as the focal point
- Emphasis on widespread training, quality awareness
- Training often turned over to HR function
- Not enough emphasis on quality control and improvement tools, poor follow-through, no project-by-project implementation strategy
- TQM was largely unsuccessful

Total Quality Management (TQM)

- TQM is “just another program”
- Value engineering
- Zero defects
- “Quality is free”

Recipe for Ineffectiveness and maybe Disaster

Quality Systems and Standards

The International Standards Organization (founded in 1946 in Geneva, Switzerland), known as ISO, has developed a series of standards for quality systems. The first standards were issued in 1987. The current version of the standard is known as the ISO 9000 series. It is a generic standard, broadly applicable to any type of organization, and it is often used to demonstrate a supplier's ability to control its processes. The three standards of ISO 9000 are:

ISO 9000:2000 Quality Management System—Fundamentals and Vocabulary

ISO 9001:2000 Quality Management System—Requirements

ISO 9004:2000 Quality Management System—Guidelines for Performance Improvement

The ISO 9001:2000 standard has eight clauses: (1) Scope, (2) Normative References, (3) Definitions, (4) Quality Management Systems, (5) Management Responsibility, (6) Resource Management, (7) Product (or Service) Realization, and (8) Measurement, Analysis, and Improvement. Clauses 4 through 8 are the most important, and their key components and requirements are shown in Table 1-2. To become certified under the ISO standard, a company must select a registrar and prepare for a certification audit by this registrar. There is no single independent authority that licenses, regulates, monitors, or qualifies registrars. As we will discuss later, this is a serious problem with the ISO system. Preparing for the certification audit involves many activities, including (usually) an initial or phase I audit that checks the present quality management system against the standard. This is usually followed by establishing teams to ensure that all components of the key clause are developed and implemented, training of personnel, developing applicable documentation, and developing and installing all new components of the quality system that may be required. Then the certification audit takes place. If the company is certified, then periodic surveillance audits by the registrar continue, usually on an annual (or perhaps six-month) schedule.

- The ISO certification process focuses heavily on quality assurance, without sufficient weight given to quality planning and quality control and improvement

There is also evidence that ISO certification or certification under one of the other industry-specific standards does little to prevent poor quality products from being designed, manufactured, and delivered to the customer. For example, in 1999–2000, there were numerous incidents of rollover accidents involving Ford Explorer vehicles equipped with Bridgestone/Firestone tires and there were nearly 300 deaths in the United States alone attributed to these accidents, which led to a recall by Bridgestone/Firestone of approximately 6.5 million tires. Apparently, many of the tires involved in these incidents were manufactured at the Bridgestone/Firestone plant in Decatur, Illinois. In an article on this story in *Time* magazine (September 18, 2000), there was a photograph (p. 38) of the sign at the entrance of the Decatur plant which stated that the plant was "QS 9000 Certified" and "ISO 14001 Certified" (ISO 14001 is an environmental standard). Although the assignable causes underlying these incidents have not been fully discovered, there are clear indicators that despite quality systems certification, Bridgestone/Firestone experienced significant quality problems.

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The Malcolm Baldrige National Quality Award

The Malcolm Baldrige National Quality Award (MBNQA) was created by the U.S. Congress in 1987. It is given annually to recognize U.S. corporations for performance excellence. Awards are given to organizations in five categories: manufacturing, service, small business, health care, and education. Three awards may be given each year in each category. Many organizations compete for the awards, and many companies use the performance excellence criteria for self-assessment. The award is administered by NIST (the National Bureau of Standards and Technology).

- The MBNQA process is a valuable assessment tool
- See Table 1-3 for Performance Excellence Criteria and point values

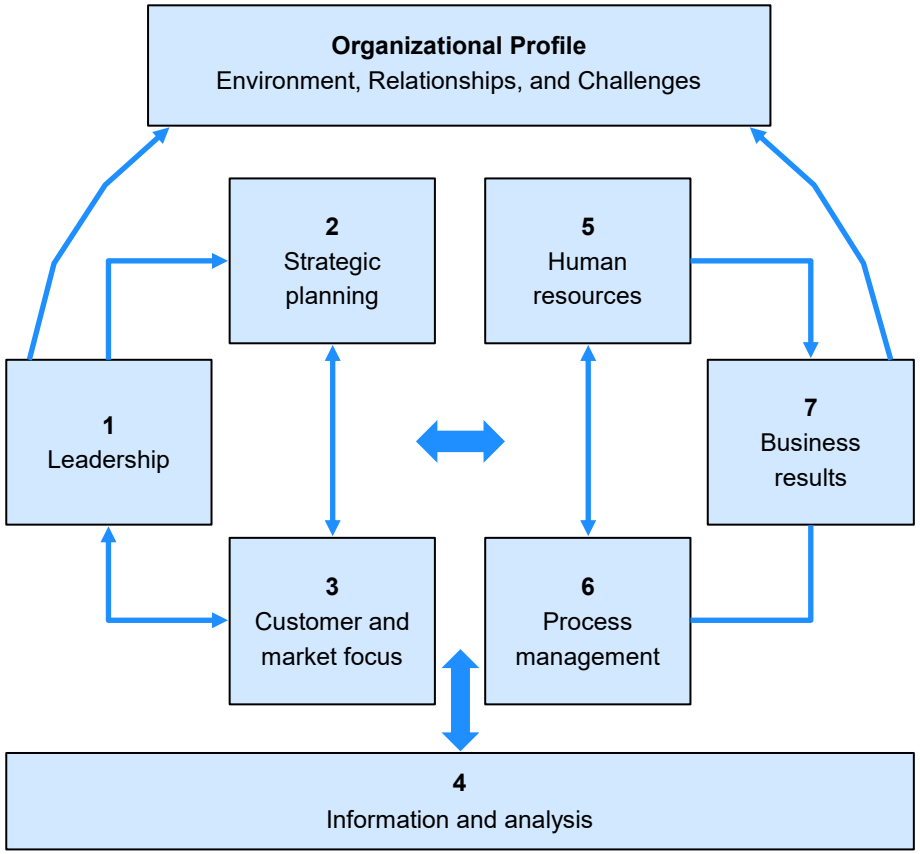


Figure 1-9 The structure of the MBNQA performance excellence criteria. (Source: Foundation for the Malcolm Baldrige Natonal Quality Award, 2002 Criteria for Performance Excellence.)

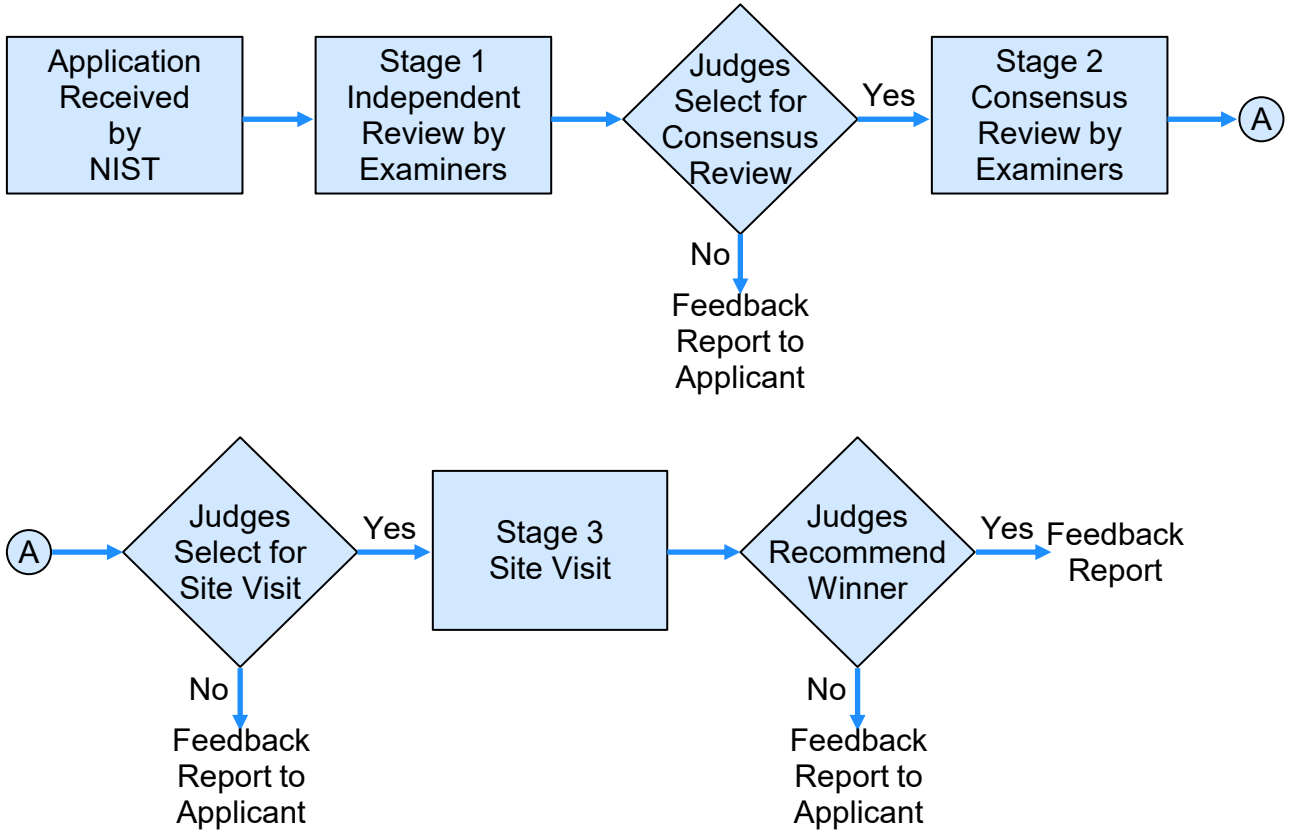


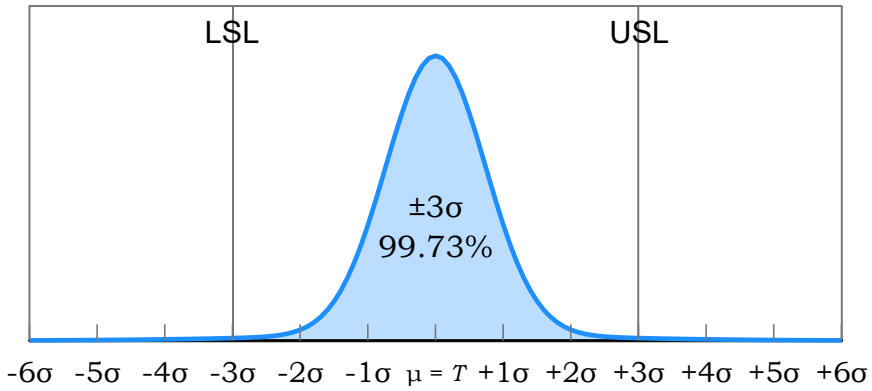
Figure 1-10 MBNQA process. (Source: Foundation for the Malcolm Baldrige Natonal Quality Award, 2002 Criteria for Performance Excellence.)

Six Sigma

- Use of statistics & other analytical tools has grown steadily for over 80 years
  - Statistical quality control (origins in 1920, explosive growth during WW II, 1950s)
  - Operations research (1940s)
  - FDA, EPA in the 1970's
  - TQM (Total Quality Management) movement in the 1980's
  - Reengineering of business processes (late 1980's)
  - Six-Sigma (origins at Motorola in 1987, expanded impact during 1990s to present)

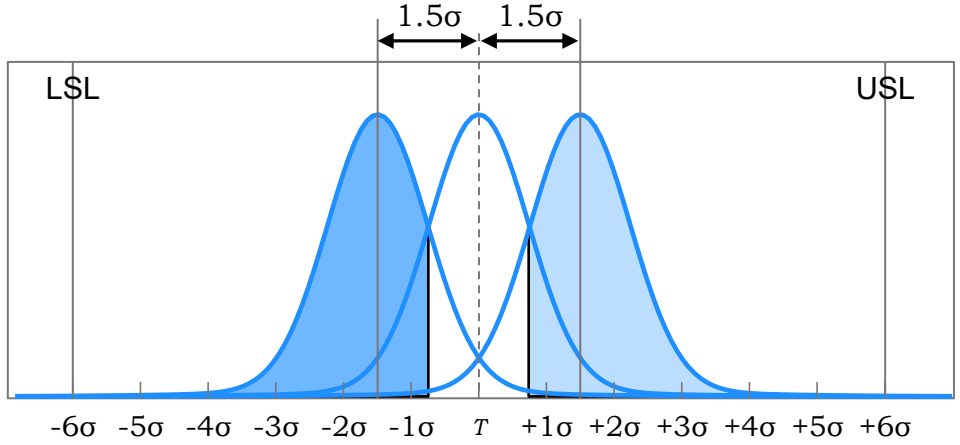
Focus of Six Sigma is on Process Improvement with an Emphasis on Achieving Significant Business Impact

- A process is an organized sequence of activities that produces an output that adds value to the organization
- All work is performed in (interconnected) processes
  - Easy to see in some situations (manufacturing)
  - Harder in others
- Any process can be improved
- An organized approach to improvement is necessary
- The process focus is essential to Six Sigma



Spec. Limit	Percent Inside Specs	ppm Defective
±1 Sigma	68.27	317300
±2 Sigma	95.45	45500
±3 Sigma	99.73	2700
±4 Sigma	99.9937	63
±5 Sigma	99.999943	0.57
±6 Sigma	99.9999998	0.002

(*a*) Normal distribution centered at the target (*T*)



Spec. Limit	Percent inside specs	ppm Defective
±1 Sigma	30.23	697700
±2 Sigma	69.13	608700
±3 Sigma	93.32	66810
±4 Sigma	99.3790	6210
±5 Sigma	99.97670	233
±6 Sigma	99.999660	3.4

(*b*) Normal distribution with the mean shifted by ±1.5σ from the target

Figure 1-11 The Motorola six-sigma concept.

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Why "Quality Improvement" is Important: A Simple Example

- A visit to a fast-food store: Hamburger (bun, meat, special sauce, cheese, pickle, onion, lettuce, tomato), fries, and drink.
- This product has 10 components - is 99% good okay?

$P\{\text{Single meal good}\} = (0.99)^{10} = 0.9044$
Family of four, once a month: $P\{\text{All meals good}\} = (0.9044)^{4} = 0.6690$
$P\{\text{All visits during the year good}\} = (0.6690)^{12} = 0.0080$

$P\{\text{single meal good}\} = (0.999)^{10} = 0.9900,\ P\{\text{Monthly visit good}\} = (0.99)^{4} = 0.9607$
$P\{\text{All visits in the year good}\} = (0.9607)^{12} = 0.6186$

Six Sigma Focus

- Initially in manufacturing
- Commercial applications
  - Banking
  - Finance
  - Public sector
  - Services
- DFSS – Design for Six Sigma
  - Only so much improvement can be wrung out of an existing system
  - New process design
  - New product design (engineering)

Some Commercial Applications

- Reducing average and variation of days outstanding on accounts receivable
- Managing costs of consultants (public accountants, lawyers)
- Skip tracing
- Credit scoring
- Closing the books (faster, less variation)
- Audit accuracy, account reconciliation
- Forecasting
- Inventory management
- Tax filing
- Payroll accuracy

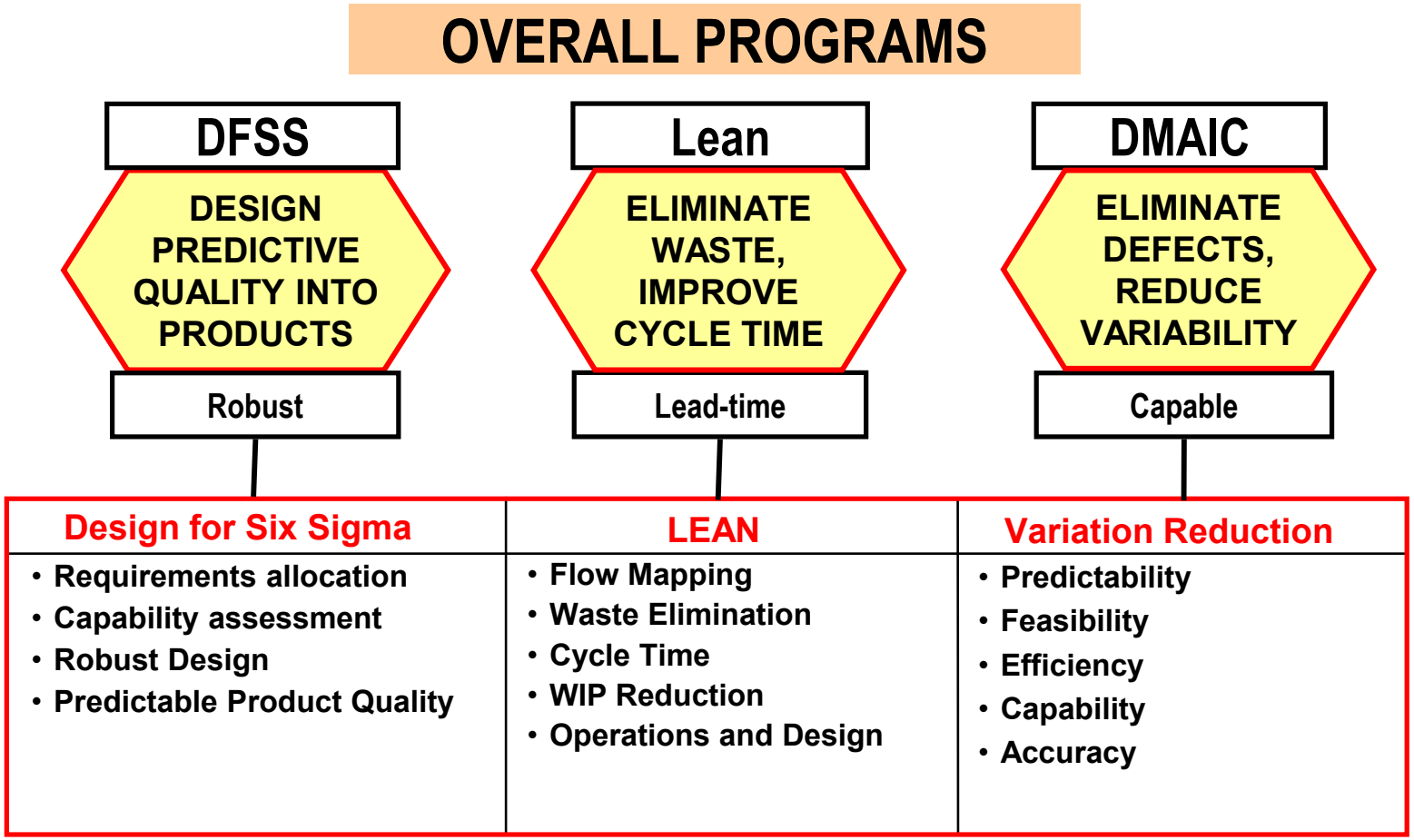
Six Sigma

- A disciplined and analytical approach to process and product improvement
- Specialized roles for people; Champions, Master Black belts, Black Belts, Green Belts
- Top-down driven (Champions from each business)
- BBs and MBBs have responsibility (project definition, leadership, training/mentoring, team facilitation)
- Involves a five-step process (DMAIC) :
  - Define
  - Measure
  - Analyze
  - Improve
  - Control

What Makes it Work?

- Successful implementations characterized by:
  - Committed leadership
  - Use of top talent
  - Supporting infrastructure
    - Formal project selection process
    - Formal project review process
    - Dedicated resources
    - Financial system integration
- Project-by-project improvement strategy (borrowed from Juran)

The Process Improvement Triad: DFSS, Lean, and DMAIC



The "I" in DMAIC may become DFSS

DFSS Matches Customer Needs with Capability

- Mean and variability affects product performance and cost
  - Designers can predict costs and yields in the design phase
- Consider mean and variability in the design phase
  - Establish top level mean, variability and failure rate targets for a design
  - Rationally allocate mean, variability, and failure rate targets to subsystem and component levels
  - Match requirements against process capability and identify gaps
  - Close gaps to optimize a producible design
  - Identify variability drivers and optimize designs or make designs robust to variability
- Process capability impact design decisions

DFSS enhances product design methods.

Lean Focuses on Waste Elimination

- Definition
- A set of methods and tools used to eliminate waste in a process
- Lean helps identify anything not absolutely required to deliver a quality product on time.
- Benefits of using Lean
- Lean methods help reduce inventory, lead time, and cost
- Lean methods increase productivity, quality, on time delivery, capacity, and sales

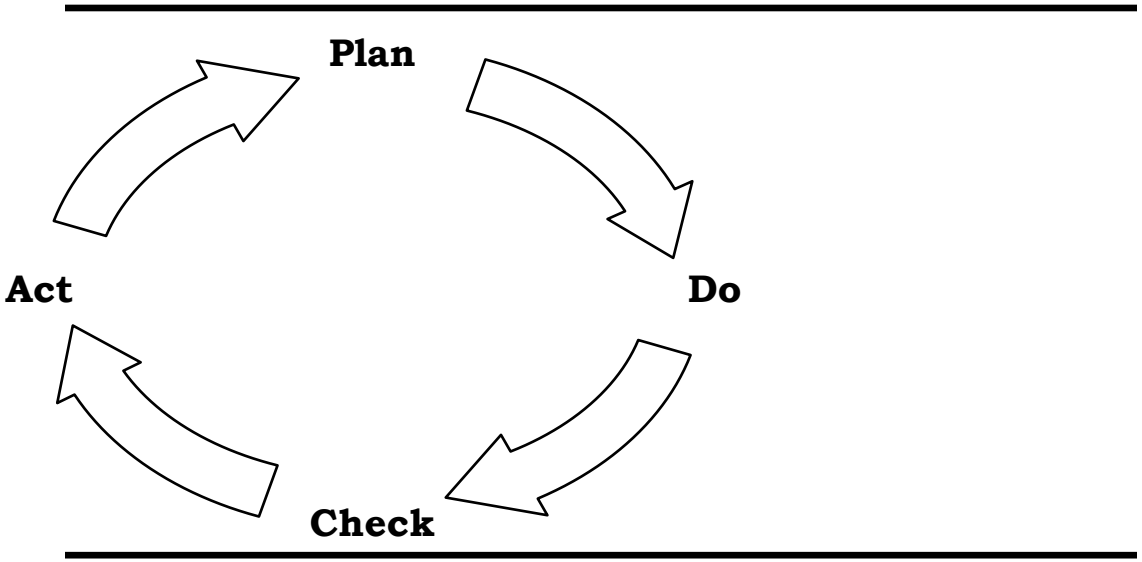
DMAIC Solves Problems by Using Six Sigma Tools

- DMAIC is a problem solving methodology

- Use this method to solve problems:
  - Define problems in processes
  - Measure performance
  - Analyze causes of problems
  - Improve processes–remove variations and nonvalue-added activities
  - Control processes so problems do not recur

Six Sigma

- DMAIC is closely related to the Shewhart cycle (variously called the Deming cycle, or the PDCA cycle)



Quality Costs

Table 1-4 Quality Costs

Prevention Costs
- Quality planning and engineering
- New products review
- Product/process design
- Process control
- Burn-in
- Training
- Quality data acquisition and analysis

Appraisal Costs
- Inspection and test of incoming material
- Product inspection and test
- Materials and services consumed
- Maintaining accuracy of test equipment

Internal Failure Costs
- Scrap
- Rework
- Retest
- Failure analysis
- Downtime
- Yield losses
- Downgrading (off-specing)

External Failure Costs
- Complaint adjustment
- Returned product/material
- Warranty charges
- Liability costs
- Indirect costs

Quality Costs

- The increase in the cost of quality because of the increase in the complexity of manufactured products associated with advances in technology.
- Increasing awareness of life cycle costs, including maintenance, labor, spare parts and the costs of field failures.
- The need for quality engineers and managers to effectively communicate the cost of quality in the language of general management – namely, money.

Quality Costs

- Prevention Costs
  - Quality planning & Engineering
  - New Products review
  - Product/process design
  - Process control
  - Burn-in
  - Training
  - Quality data acquisition & analysis
- Internal Failure costs
  - Scrap
  - Rework
  - Retest
  - Failure analysis
  - Downtime
  - Yield losses
  - Downgrading (off-specing)

Quality Costs

- Appraisal Costs
  - Inspection and test of incoming
  - Material
  - Product inspection & test
  - Materials & services consumed
  - Maintaining accuracy of test equipment
- External Failure costs
  - Complaint adjustment
  - Returned product / material
  - Warranty charges
  - Liability costs
  - Indirect costs

Legal Aspects of Quality

- Product liability exposure
- Concept of **strict liability**
  1. Responsibility of both manufacturer and seller/distributor
  2. Advertising must be supported by valid data

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Implementing Quality Improvement

- A strategic management process, focused along the eight dimension of quality

Table 1-6 The Eight Dimensions of Quality from Section 1-1.1

1. Performance
2. Reliability
3. Durability
4. Serviceability
5. Aesthetics
6. Features
7. Perceived quality
8. Conformance to standards

- Suppliers and supply chain management must be involved
- Must focus on all three components: Quality Planning, Quality Assurance, and Quality Control and Improvement

IMPORTANT TERMS AND CONCEPTS

Acceptance sampling
Appraisal costs
Deming's 14 points
Designed experiments
Dimensions of quality
Fitness for use
Internal and external failure costs
ISI 9000:2000
Nonconforming product or service
Prevention costs
Product liability
Quality assurance
Quality characteristics
Quality control and improvement
Quality engineering
Quality of conformance
Quality of design
Quality planning
Quality systems and standards
Six-Sigma
Specifications
Statistical process control (SPC)
The Juran Trilogy
The Malcolm Baldrige National Quality Award
Total quality management (TQM)
Variability

Learning Objectives

1. Define and discuss quality and quality improvement
2. Discuss the different dimensions of quality
3. Discuss the evolution of modern quality improvement methods
4. Discuss the role that variability and statistical methods play in controlling and improving quality
5. Describe the quality management philosophies of W. Edwards Deming, Joseph M. Juran, and Armand V. Feigenbaum
6. Discuss total quality management, the Malcolm Baldrige National Quality Award, Six-Sigma, and quality systems and standards
7. Explain the links between quality and productivity and between quality and cost
8. Discuss product liability
9. Discuss the three functions: quality planning, quality assurance, and quality control and improvement

Methods and Philosophy of Statistical Process Control

Chapter 4. Methods and Philosophy of Statistical Process Control

Basic SPC Tools

SPC can be applied to *any* process. Its seven major tools are

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

- In statistical control: a process operating with only chance causes of variation
- Out of control: a process operating in the presence of assignable causes

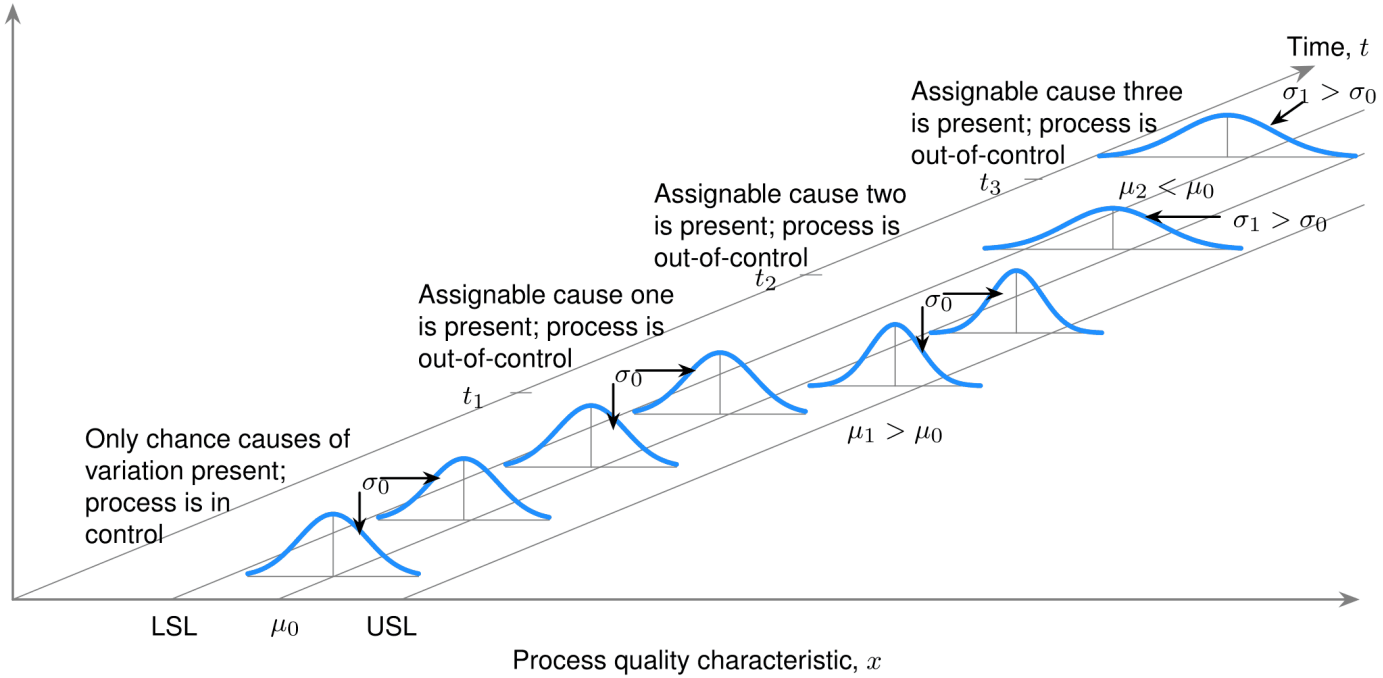


Figure 4-1 Chance and assignable causes of variation.

- A control chart contains
  - A **center line**
  - An **upper control limit**
  - A **lower control limit**
- A point that plots within the control limits indicates the process is in control
- A point that plots outside the control limits is evidence that the process is out of control
- There is a close connection between control charts and hypothesis testing

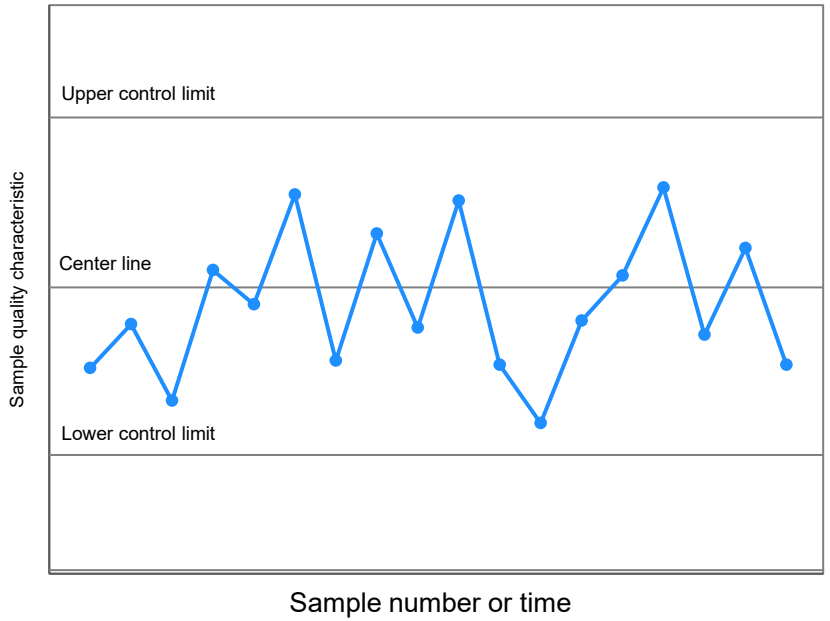


Figure 4-2 A typical control chart.

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Photolithography Example

- Important quality characteristic in hard bake is resist flow width
- Process is monitored by average flow width
  - Sample of 5 wafers
  - Process mean is 1.5 microns
  - Process standard deviation is 0.15 microns
- Note that all plotted points fall inside the control limits
  - Process is considered to be in statistical control

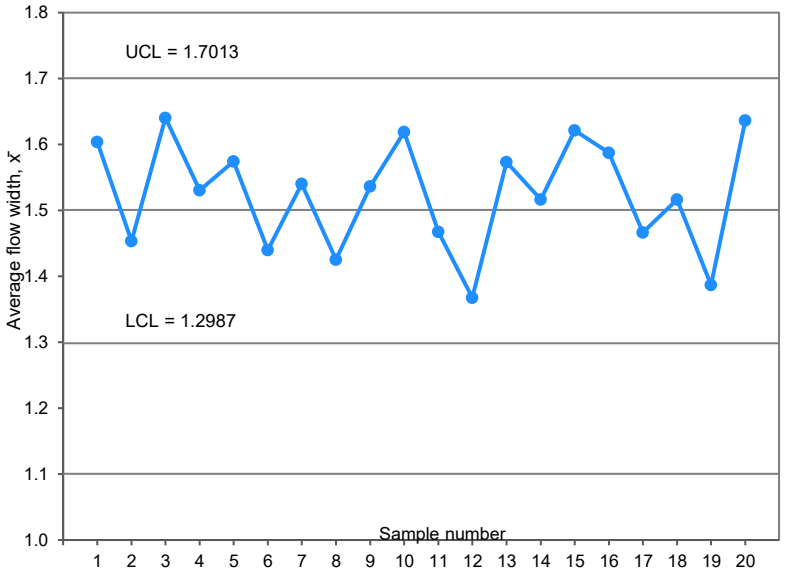


Figure 4-3 $\bar{x}$ control chart for flow width.

The process mean is 1.5 microns, and the process standard deviation is σ = 0.15 microns. Now if samples of size n = 5 are taken, the standard deviation of the sample average $\overline{x}$ is

$$\sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.15}{\sqrt{5}} = 0.0671$$

Therefore, if the process is in control with a mean flow width of 1.5 microns, then by using the central limit theorem to assume that $\overline{x}$ is approximately normally distributed, we would expect $100(1 - \alpha)\%$ of the sample means $\overline{x}$ to fall between $1.5 + Z_{\alpha/2}(0.0671)$ and $1.5 - Z_{\alpha/2}(0.0671)$. We will arbitrarily choose the constant $Z_{\alpha/2}$ to be 3, so that the upper and lower control limits become

$$\text{UCL} = 1.5 + 3(0.0671) = 1.7013$$

and

$$\text{LCL} = 1.5 - 3(0.0671) = 1.2987$$

as shown on the control chart. These are typically called **"three-sigma"**[2] **control limits.**

Shewhart Control Chart Model

w: sample variable
μ_w: mean of w
δ_{*w*}: standard deviation of w
L: distance in terms of δ_{*w*}

$$\text{UCL} = \mu_w + L\sigma_w$$
$$\text{Center line} = \mu_w$$
$$\text{LCL} = \mu_w - L\sigma_w \quad (4\text{-}1)$$

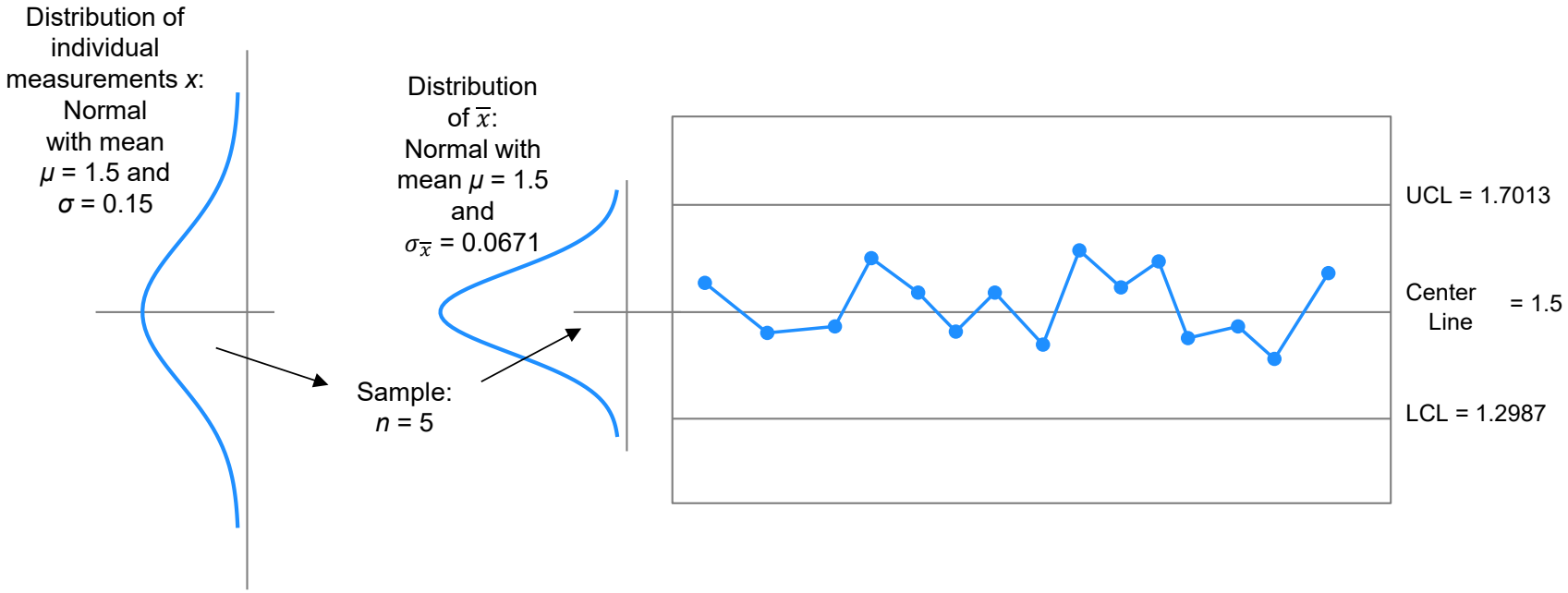


Figure 4-4 How the control chart works.

Improving Quality

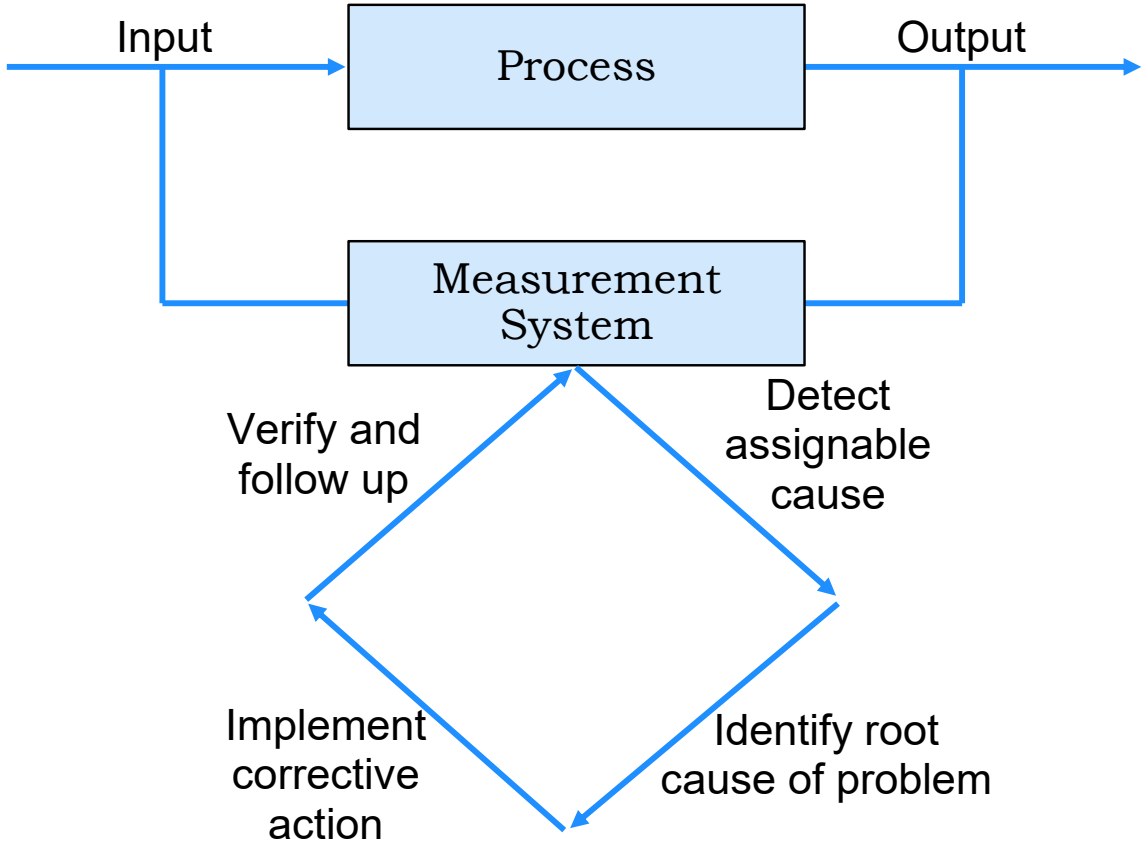
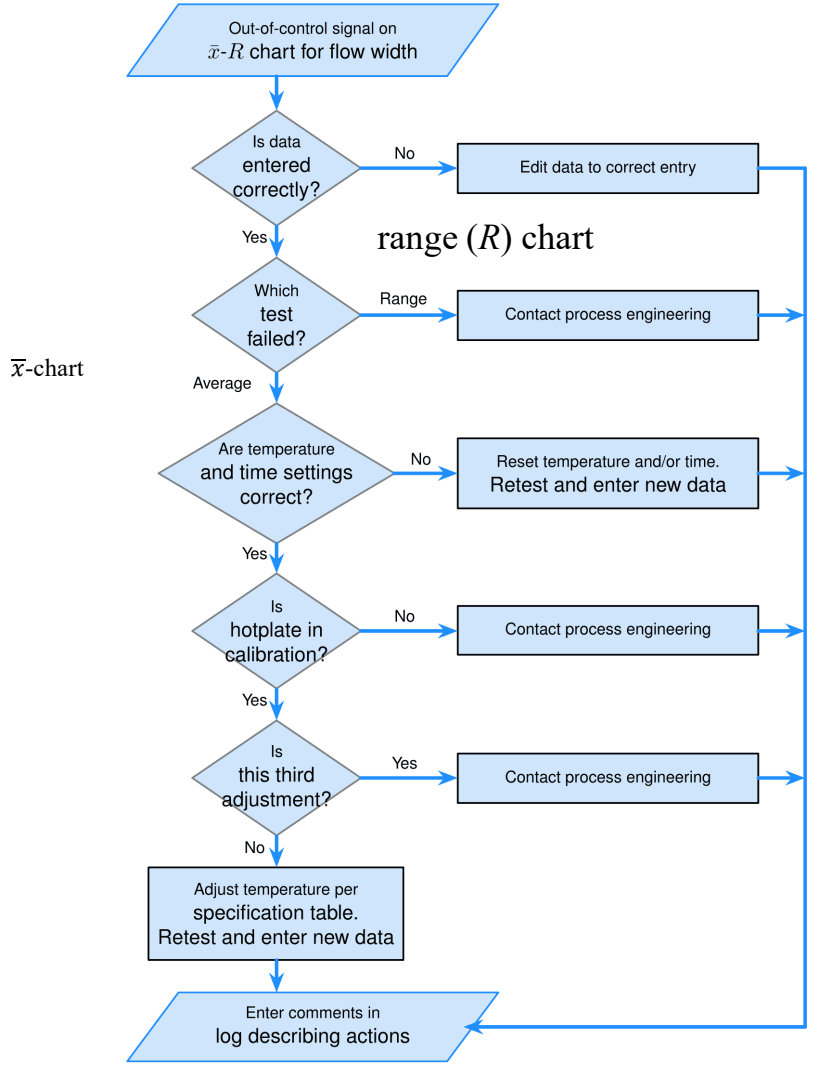


Figure 4-5 Process improvement using the control chart.

Out of Control Action Plan (OCAP)



More Basic Principles

- Charts may be used to estimate process parameters, which are used to determine capability
- Two general types of control charts
  - Variables (Chapter 5)
    - Continuous scale of measurement
    - Quality characteristic described by central tendency and a measure of variability
  - Attributes (Chapter 6)
    - Conforming/nonconforming
    - Counts
- Control chart design encompasses selection of sample size, control limits, and sampling frequency

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Types of Process Variability

- Stationary and uncorrelated – data vary around a fixed mean in a stable or predictable manner
- Stationary and autocorrelated – successive observations are dependent with tendency to move in long runs on either side of mean
- Nonstationary – process drifts without any sense of a stable or fixed mean

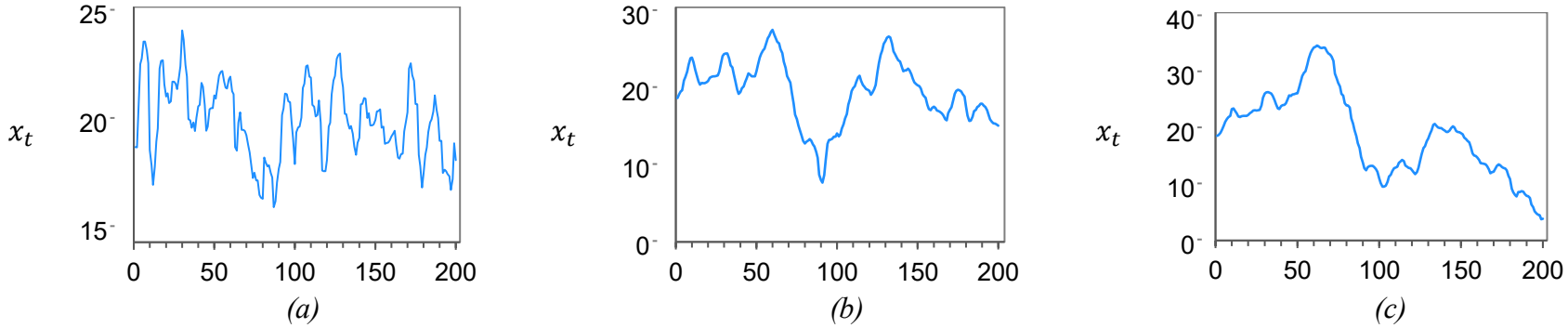


Figure 4-7 Data from three different processes. (*a*) Stationary and uncorrelated (white noise). (*b*) Stationary and autocorrelated. (*c*) Nonstationary.

Reasons for Popularity of Control Charts

1. Control charts are a proven technique for improving productivity.
2. Control charts are effective in defect prevention.
3. Control charts prevent unnecessary process adjustment.
4. Control charts provide diagnostic information.
5. Control charts provide information about process capability.

- 3-Sigma Control Limits
  - Probability of type I error is 0.0027
- Probability Limits
  - Type I error probability is chosen directly
  - For example, 0.001 gives 3.09-sigma control limits
- Warning Limits
  - Typically selected as 2-sigma limits

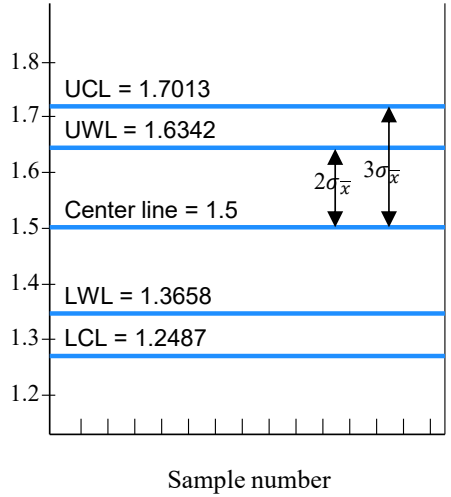


Figure 4-8 An $\bar{x}$ chart with two-sigma warning limits.

Choice of Control Limits

- By moving the control limits farther from the center line, the risk of type I error decreases and the risk of type II error increases.
- If the control limits are moved closer to the center line, the risk of type I error increases and the risk of type II error decreases.

Type I Error: is the risk of a point falling beyond the control limits indicating an out-of-control condition when no assignable cause is present.

Type II Error: is the risk of a point falling between the control limits when the process is really out of control.

Sample Size and Sampling Frequency

Average Run Length (ARL): Average number of points plotted before a point indicates an out of control condition

p: probability any point exceeds control limits

$$\text{ARL} = \frac{1}{p} \tag{4-2}$$

Average time to signal (ATS)

h: in hours

$$\text{ATS} = \text{ARL}\, h \tag{4-3}$$

Rational Subgroups

Rational Subgroups

- Rational subgroup: subgroups or samples should be selected so that if assignable causes are present, chance for differences *between* subgroups will be maximized, while chance for difference due to assignable causes *within* a subgroup will be minimized.
- Two general approaches
  1. Sample produced at the same time consecutive units
  2. Sample representing all units produced since last sample
     - Often used to make decisions about acceptance of product
     - Effective at detecting shifts to out-of-control state and back into in-control state *between* samples
     - Care must be taken because we can often make any process appear to be in statistical control just by stretching out the interval between observations in the sample.

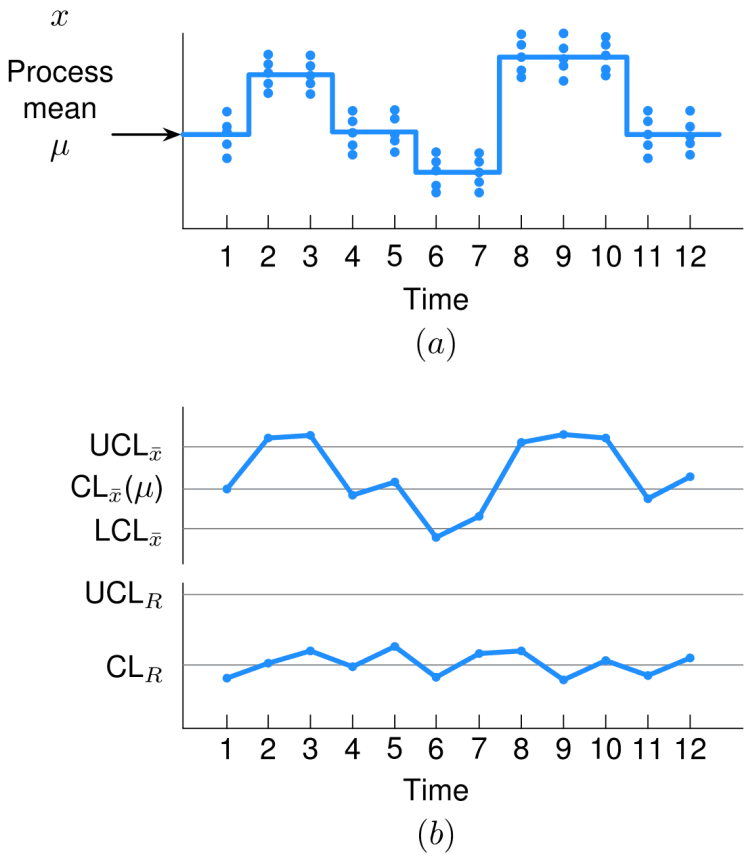


Figure 4-10 The "snapshot" approach to rational subgroups. (a) Behavior of the process mean. (b) Corresponding $\bar{x}$ and R control charts.

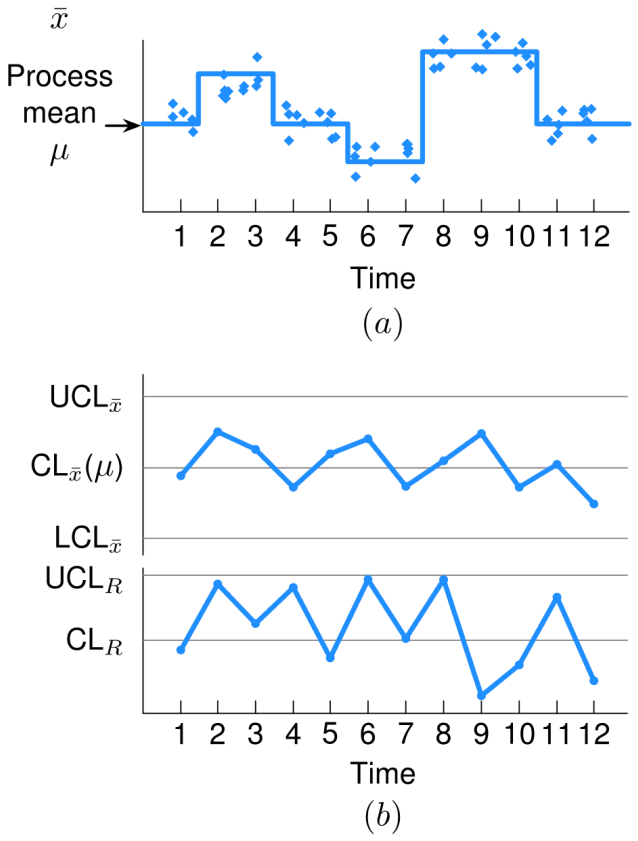


Figure 4-11 The random sample approach to rational subgroups. (a) Behavior of the process mean. (b) Corresponding $\bar{x}$ and R control charts.

Analysis of Patterns on Control Charts

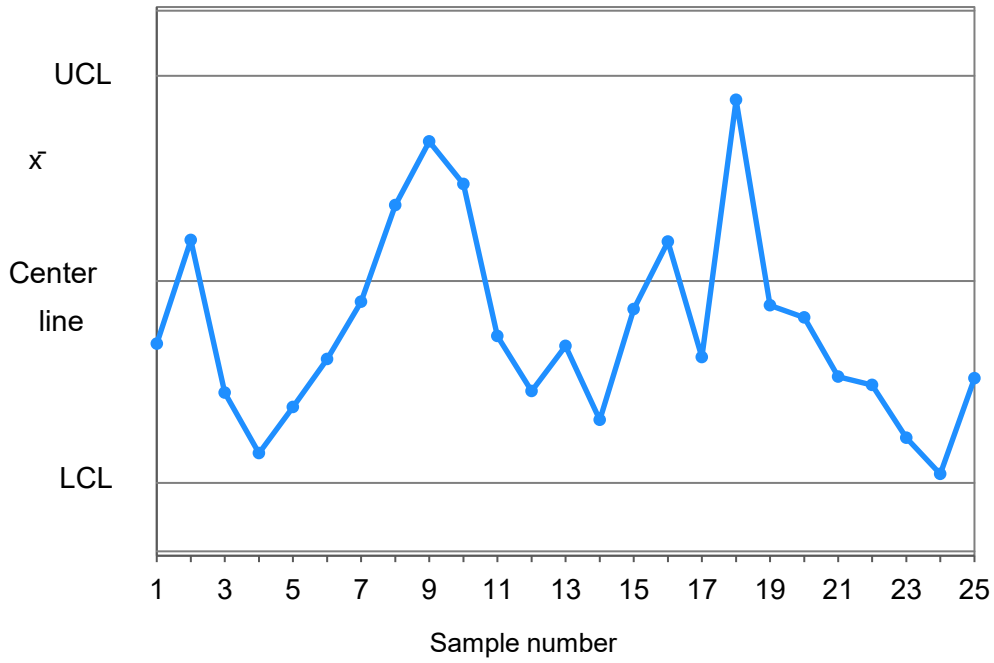


Figure 4-12 An $\bar{x}$ control chart.

- Pattern is very nonrandom in appearance
  - 19 of 25 points plot below the center line, while only 6 plot above
  - Following 4th point, 5 points in a row increase in magnitude, a *run up*
  - There is also an unusually long *run down* beginning with 18th point

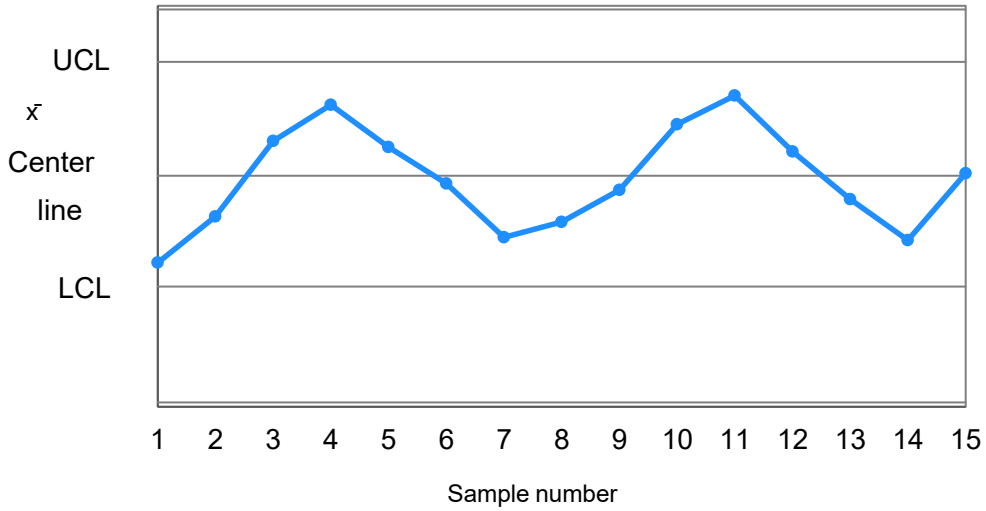


Figure 4-13 An $\bar{x}$ chart with a cyclic pattern.

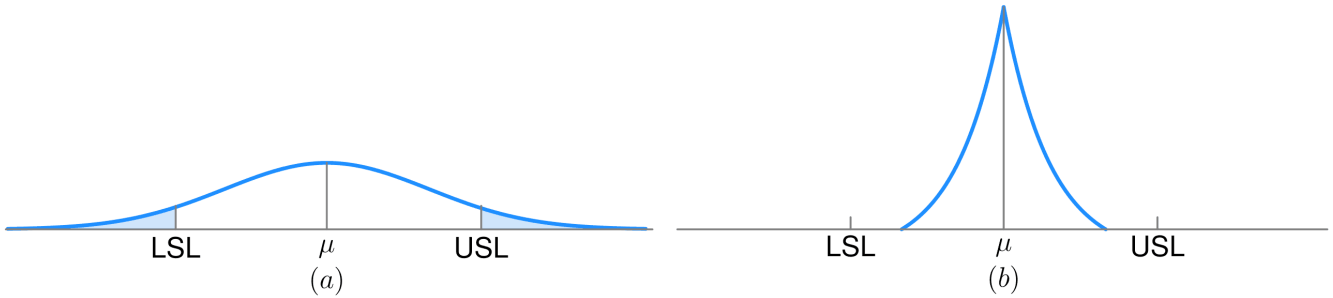


Figure 4-14 (*a*) Variability with the cyclic pattern. (*b*) Variability with the cyclic pattern eliminated.

Western Electric Rule for Out of Control

2. Two out of three consecutive points plot beyond the two-sigma warning limits;
3. Four out of five consecutive points plot at a distance of one-sigma or beyond from the center line;

or

4. Eight consecutive points plot on one side of the center line.

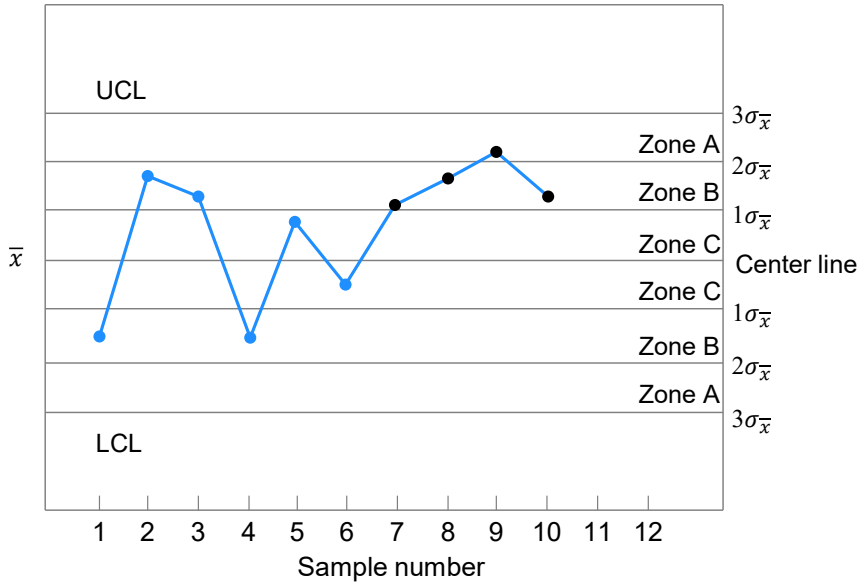


Figure 4-15 The Western Electric or zone rules, with the last four points showing a violation of the rule 3.

Additional Rules (Sensitizing Rules)

Table 4-1 Some Sensitizing Rules for Shewhart Control Charts

Standard Action Signal:

1. One or more points outside of the control limits.
2. Two of three consecutive points outside the two-sigma warning limits but still inside the control limits.
3. Four of five consecutive points beyond the one-sigma limits.
4. A run of eight consecutive points on one side of the center line.
5. Six points in a row steadily increasing or decreasing.
6. Fifteen points in a row in zone C (both above and below the center line).
7. Fourteen points in a row alternating up and down.
8. Eight points in a row on both sides of the center line with none in zone C.
9. An unusual or nonrandom pattern in the data.
10. One or more points near a warning or control limit.

Western Electric Rules

Assume k decision rules.

α_i = probability of type I error under rule i

$$\alpha = 1 - \prod_{i=1}^{k}(1 - \alpha_i) \quad (4\text{-}4)$$

α = overall type I error probability when k decision rules are independent.

Phase I and Phase II of Control Chart Application

- Phase I: Retrospective analysis of process data to construct trial control limits
  - Effective at detecting large, sustained shifts in process parameters, outliers, measurement errors, data entry errors, etc.
  - Facilitates identification and removal of assignable causes
- In phase II: Process monitoring
  - Process assumed to be reasonably stable
  - Emphasis on process monitoring, not on bringing an unruly process into control

THE "MAGNIFICENT SEVEN"

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart

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Check Sheet

CHECK SHEET
DEFECT DATA FOR 2002–2003 YTD

Part No.: TAX-41 Location: Bellevue Study Date: 6/5/03 Analyst: TCB

	2002												2003					
Defect	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**	**4**	**5**	**Total**
Parts damaged		1		3	1	2		1		10	3		2	2	7	2		34
Machining problems			3	3				1	8		3		8	3				29
Supplied parts rusted			1	1		2	9											13
Masking insufficient		3	6	4	3	1												17
Misaligned weld	2																	2
Processing out of order	2															2		4
Wrong part issued		1						2										3
Unfinished fairing			3															3
Adhesive failure				1							1		2			1	1	6
Powdery alodine					1													1

Check Sheet (cont'd)

	2002												2003					
Defect	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**1**	**2**	**3**	**4**	**5**	**Total**
Paint out of limits						1								1				2
Paint damaged by etching			1															1
Film on parts						3		1	1									5
Primer cans damaged								1										1
Voids in casting									1	1								2
Delaminated composite										2								2
Incorrect dimensions											13	7	13	1		1	1	36
Improper test procedure										1								1
Salt-spray failure													4			2		4
TOTAL	**4**	**5**	**14**	**12**	**5**	**9**	**9**	**6**	**10**	**14**	**20**	**7**	**29**	**7**	**7**	**6**	**2**	**166**

Figure 4-16 A check sheet to record defects on a tank used in an aerospace application.

Pareto Chart

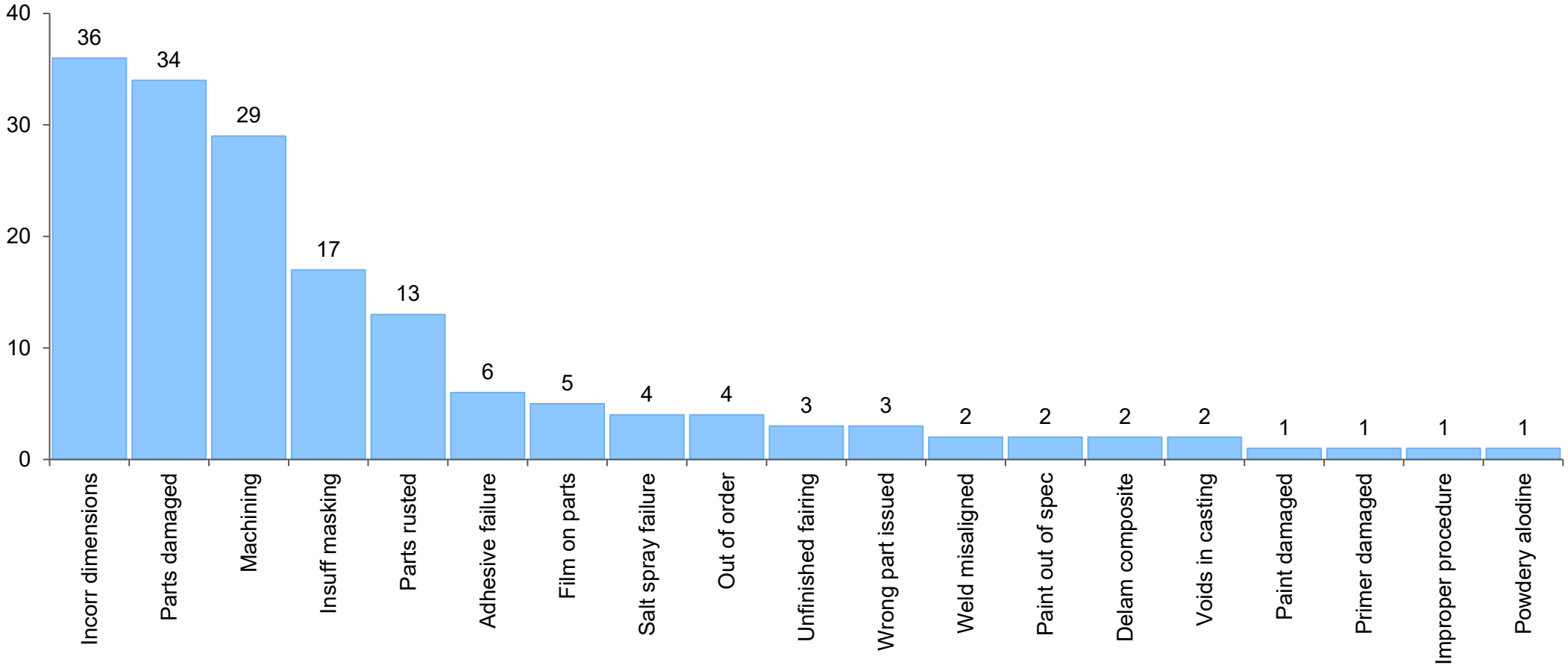


Figure 4-17 Pareto chart of the tank defect data.

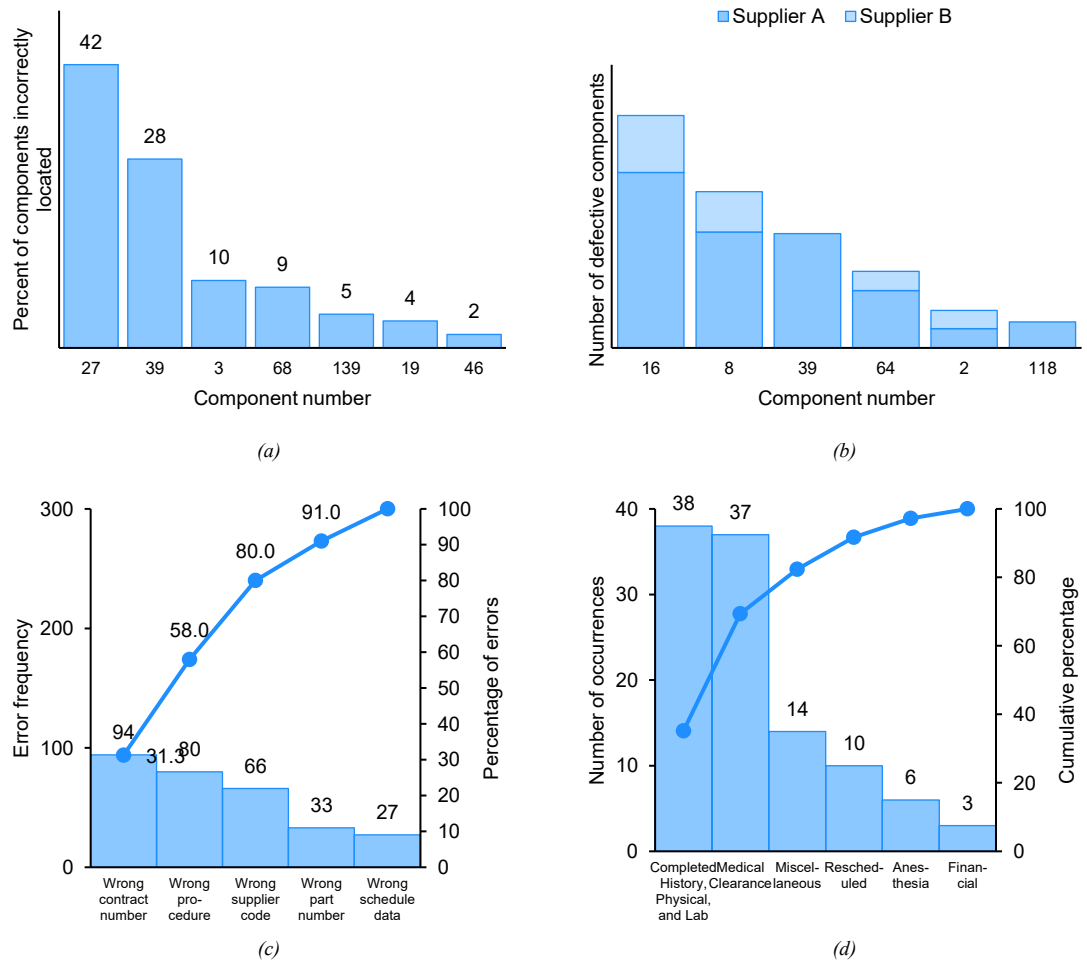


Figure 4-18 Various examples of Pareto charts.

Cause-and-Effect Diagram

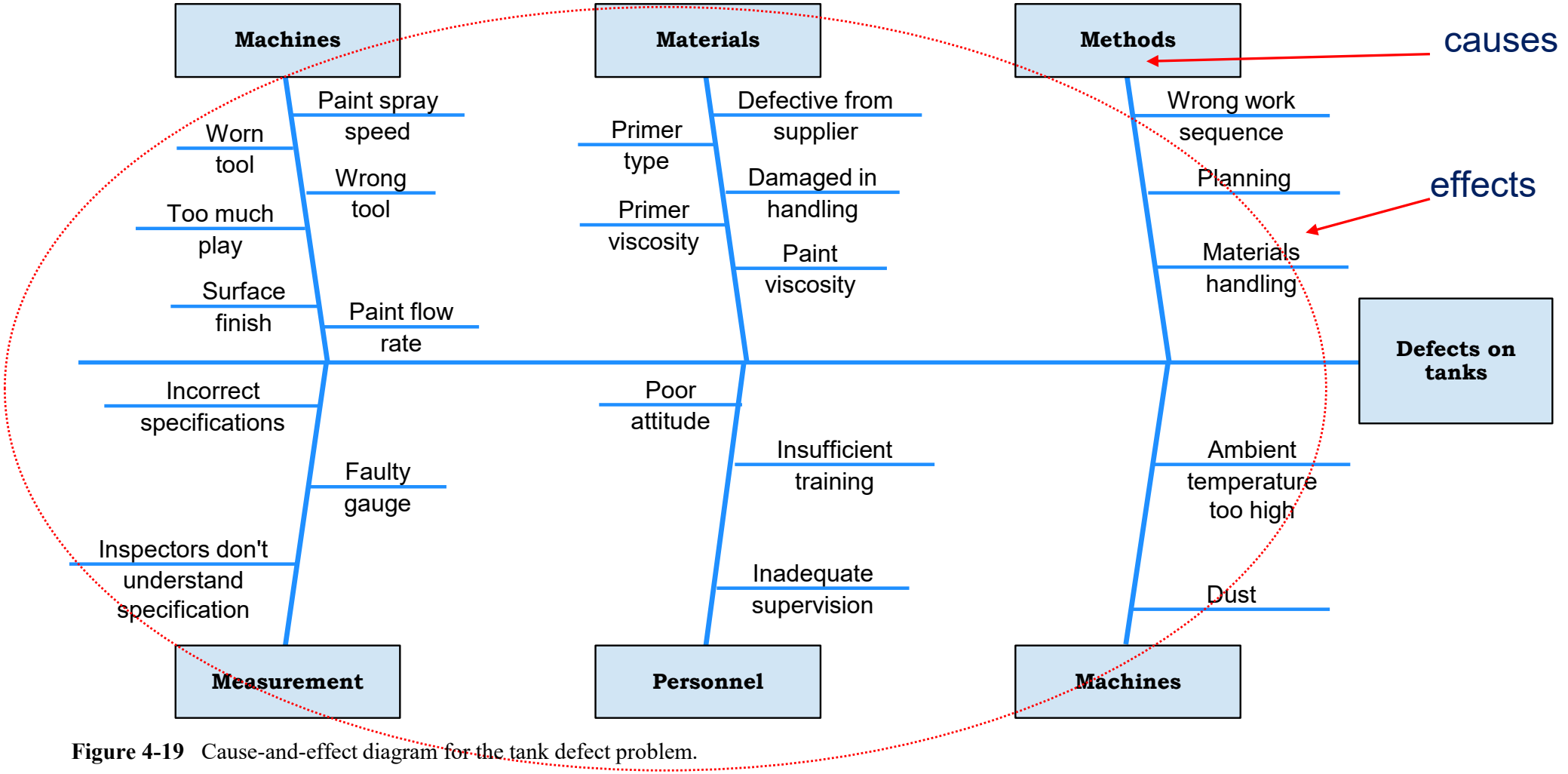


Figure 4-19 Cause-and-effect diagram for the tank defect problem.

How to Construct a Cause-and-Effect Diagram

1. Define the problem or effect to be analyzed.
2. Form the team to perform the analysis. Often the team will uncover potential causes through brainstorming.
3. Draw the effect box and the center line.
4. Specify the major potential cause categories and join them as boxes connected to the center line.
5. Identify the possible causes and classify them into the categories in step 4. Create new categories, if necessary.
6. Rank order the causes to identify those that seem most likely to impact the problem.
7. Take corrective action.

Defect Concentration Diagram

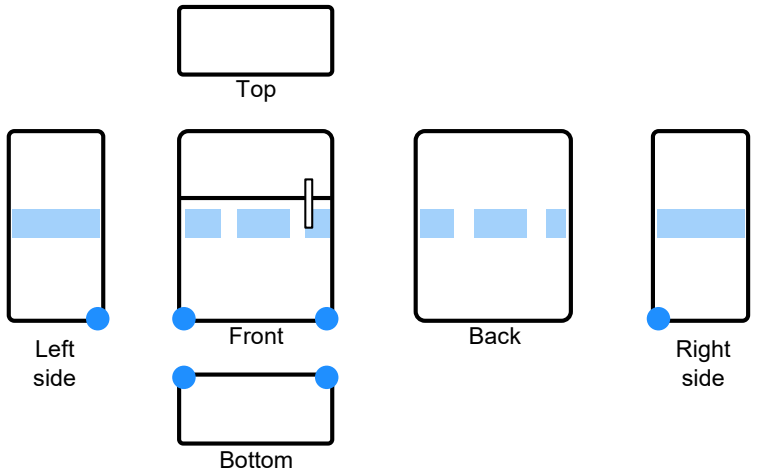


Figure 4-20 Surface-finish defects on a refrigerator.

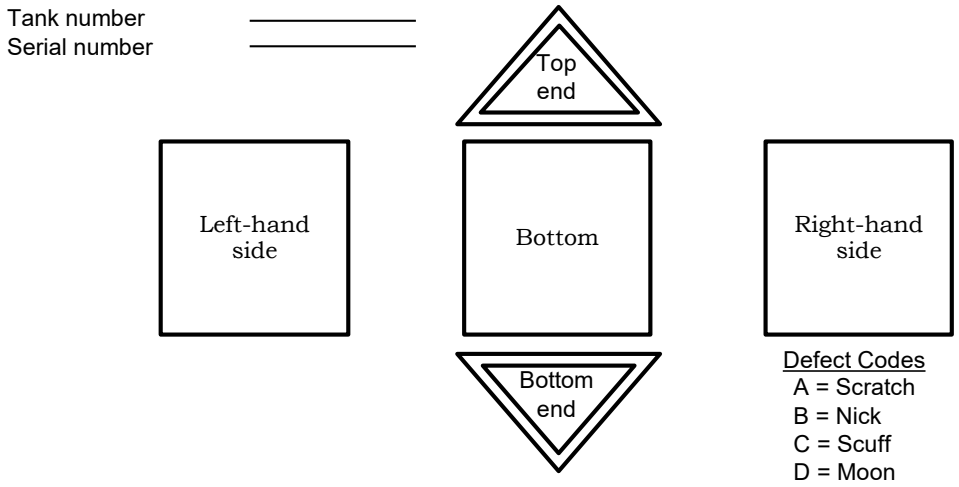


Figure 4-21 Defect concentration diagram for the tank.

Scatter Diagram

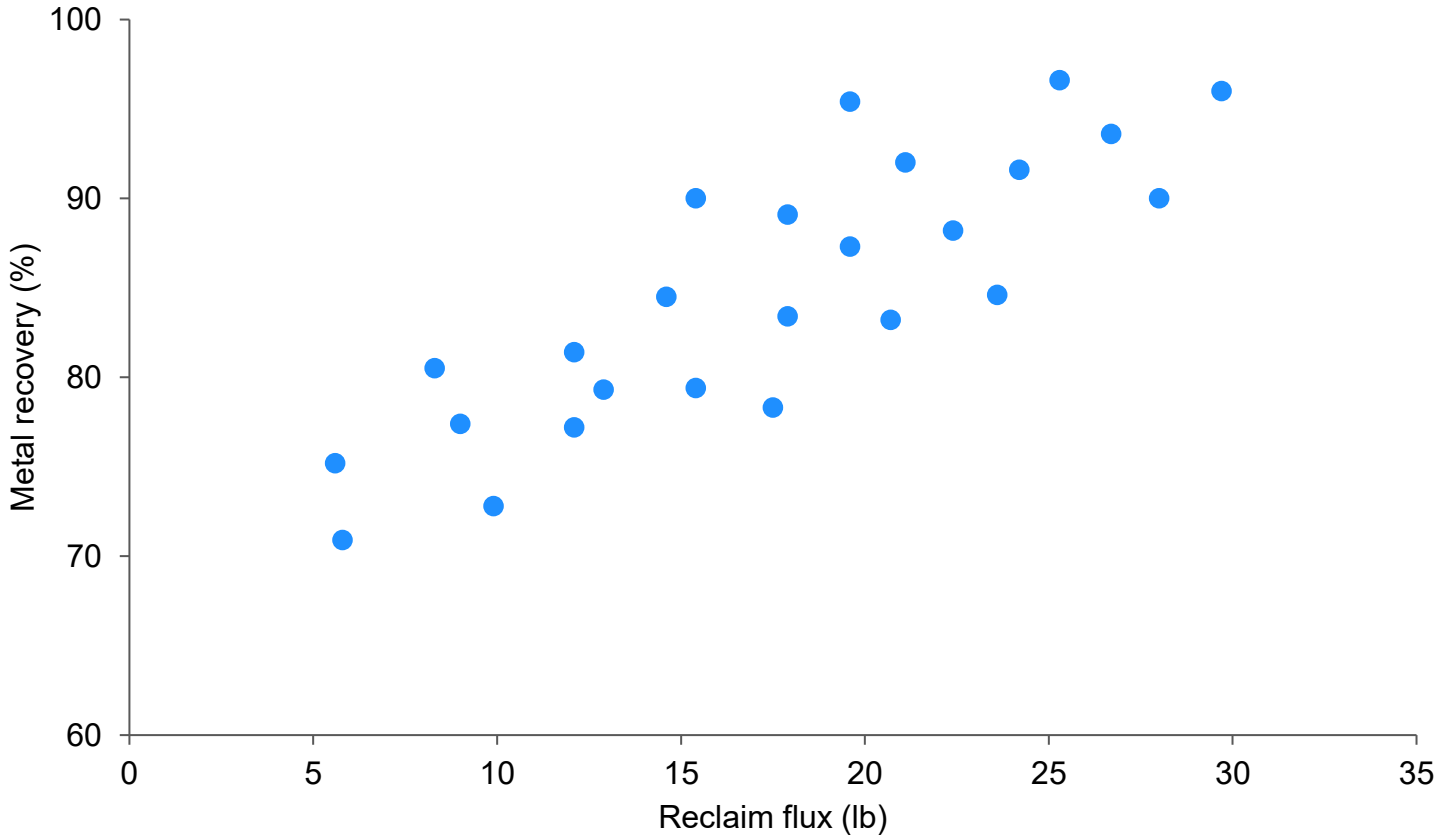


Figure 4-22 A scatter diagram.

Implementing SPC

Elements of a Successful SPC Program

1. Management leadership
2. A team approach
3. Education of employees at all levels
4. Emphasis on reducing variability
5. Measuring success in quantitative (economic) terms
6. A mechanism for communicating successful results throughout the organization

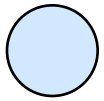
Nonmanufacturing Application of SPC

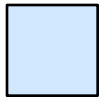
- Nonmanufacturing applications do not differ substantially from industrial applications, but sometimes require ingenuity
  1. Most nonmanufacturing operations do not have a natural measurement system
  2. The observability of the process may be fairly low
- Flow charts and operation process charts are particularly useful in developing process definition and process understanding.
  - Used to identify value-added versus nonvalue-added activity

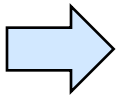
Ways to Eliminate Nonvalue-Add Activities

1. Rearranging the sequence of worksteps
2. Rearranging the physical location of the operator in the system
3. Changing work methods
4. Changing the type of equipment used in the process
5. Redesigning forms and documents for more efficient use
6. Improving operator training
7. Improving supervision
8. Identifying more clearly the function of the process to all employees
9. Trying to eliminate unnecessary steps
10. Trying to consolidate process steps

Operation Process Chart Symbols

 = operation

 = inspection

 = movement or transportation

D = delay

 = storage

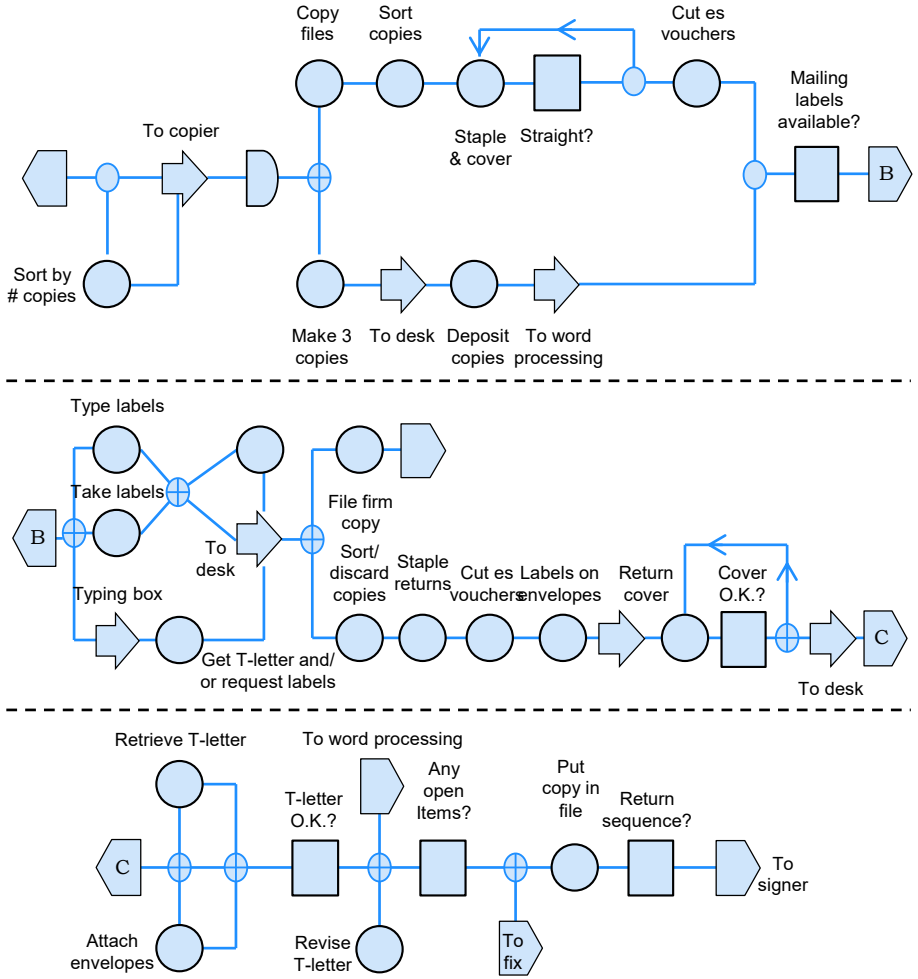


Figure 4-31 Flow chart of the assembly portion of the Form 1040 tax return process.

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Quality control measurements

- Variable – Continuous data that is concerned with degree of conformance to specifications.
- Generally expressed with statistical measures such as *averages and standard deviations*.
- Sophisticated instruments (caliper) used.
- In statistical sense, attributes inspection less efficient than variable inspection.
- Attribute data requires larger sample than variable inspection to obtain same amount of statistical information.
- Most quality characteristics in service industry are attributes.

Quality control measurements

- Attributes – A performance characteristics *that is either present or absent* in the product or service under consideration.
- Examples: Order is either complete or incomplete; an invoice can have one, two, or more errors.
- Attributes data are *discrete and tell whether the characteristics conforms to specifications*.
- Attributes measurements typically represented as proportions or rates. e.g. rate of errors per opportunity.
- Typically measured by “Go-No Go” gauges.


Thank You!